

SAMSUNG

MICROWAVE OVEN

BASIC MODEL : ME11A7510*/AA
ME11A7510*/AC

MODEL CODE : ME11CB7510*AA
ME11CB7510*AC

SERVICE *Manual*

MICROWAVE OVEN



CONTENTS

1. Precaution
2. Specifications
3. Disassembly and Reassembly
4. Alignment and Adjustments
5. Troubleshooting
6. PCB Diagrams
7. Wiring Diagrams

• Contents

1. Precaution	4
1-1 Safety precautions	5
1-2 Special High Voltage Precautions	6
2. Specifications	7
2-1 Features	7
2-2 Table of Specifications	8
2-3 Accessory	10
3. Disassembly and Reassembly	11
3-1 Tool for assembly and disassembly	11
3-2 Replacement of High Voltage Transformer	12
3-3 Replacement of Magnetron	14
3-4 Replacement of Fan Motor	14
3-5 Replacement of Door Assembly	16
3-6 Replacement of Fuse	18
3-7 Replacement of Drive Motor	18
3-8 Replacement of Control Circuit Board	19
3-9 Replacement of Cooktop Lamp	21
3-10 Replacement of Oven Light	22
4. Alignment and Adjustments	23
4-1 High Voltage Transformer	23
4-2 Magnetron	23
4-3 High Voltage Capacitor	23
4-4 High Voltage Diode	24
4-5 Main Relay and Power Control Relay	24
4-6 Adjustment of Primary interlock switch, Door sensing switch (Secondary interlock switch), and Monitor interlock switch	24
4-7 Vent Exhaust Blower Motor	25
4-8 Thermal Cutout (TCO'S)	26
4-9 Output Power of Magnetron	28
4-10 Procedure for Measurement of Microwave Energy Leakage	29
4-11 Check for Microwave Leakage	29
4-12 Note in Measurement	29
4-13 Leakage Measuring Procedure	29
4-14 Sensor	30

• Contents

5. Troubleshooting	31
5-1 Electrical Malfunction	31
5-2 Information Code Numbering Rule	34
5-3 Information Code List	34
6. PCB Diagrams	38
6-1 PCB Diagrams (Sub Module)	38
6-2 PCB Diagrams (Main PBA)	40
7. Wiring Diagrams	42
7-1 Wiring Diagrams	42

1. Precaution

PRECAUTIONS TO BE OBSERVED BEFORE AND DURING SERVICING TO AVOID POSSIBLE EXPOSURE TO EXCESSIVE MICROWAVE ENERGY

- (a) Do not operate or allow the oven to be operated with the door open.
- (b) Make the following safety checks on all ovens to be serviced before activating the magnetron or other microwave source, and make repairs as necessary:
 - (1) Interlock operation,
 - (2) Proper door closing
 - (3) Seal and sealing surfaces
(arcing, wear, and other damage),
 - (4) Damage to or loosening of hinges and latches,
 - (5) Evidence of dropping or abuse.
- (c) Before turning on microwave power for any service test or inspection within the microwave generating compartments, check the magnetron, wave guide or transmission line, and cavity for proper alignment, integrity, and connections.
- (d) Any defective or misadjusted components in the interlock, monitor, door seal, and microwave generation and transmission systems shall be repaired, replaced, or adjusted by procedures described in this manual before the oven is released to the owner.
- (e) A Microwave leakage check to verify compliance with the Federal performance standard should be performed on each oven prior to release to the owner.

1. Precaution

Follow these special safety precautions. Although the microwave oven is completely safe during ordinary use, repair work can be extremely hazardous due to possible exposure to microwave radiation, as well as potentially lethal high voltages and currents.

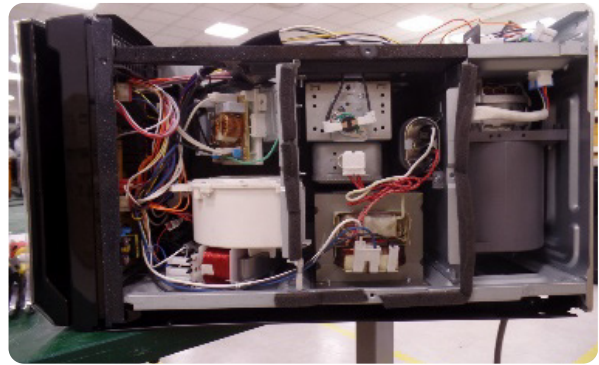
1-1 Safety precautions ()

1. All repairs should be done in accordance with the procedures described in this manual. This product complies with Federal Performance Standard 21 CFR
 2. Microwave emission check should be performed prior to servicing if the oven is operative.
 3. If the oven operates with the door open :Instruct the user not to operate the oven and contact the manufacturer and the center for devices and radiological health immediately.
 4. Notify the Central Service Center if the microwave leakage exceeds 5 mW/cm².
 5. Check all grounds.
 6. Do not power the MWO from a "2-prong" AC cord. Be sure that all of the built-in protective devices are replaced. Restore any missing protective shields.
 7. When reinstalling the chassis and its assemblies, be sure to restore all protective devices, including: nonmetallic control knobs and compartment covers.
 8. Make sure that there are no cabinet openings through which people --particularly children--might insert objects and contact dangerous voltages. Examples: Lamp hole,ventilation slots.
 9. Inform the manufacturer of any oven found to have emission in excess of 5 mW/cm² ,make repairs to bring the unit into compliance at no cost to owner and try to determine cause. Instruct owner not to use oven until it has been brought into compliance.
- CENTRAL SERVICE CENTER**
10. Service technicians should remove their watches while repairing an MWO.
 11. To avoid any possible radiation hazard,replace parts in accordance with the wiring diagram. Also, use only the exact replacements for the following parts: Primary and secondary interlock switches, interlock monitor switch.
 12. If the fuse is blown by the Interlock Monitor Switch: Replace all of the following at the same time: Primary, door sensing switch and power relay, as well as the Interlock Monitor Switch. The correct adjustment of these switches is described elsewhere in this manual. Make sure that the fuse has the correct rating for the particular model being repaired.
 13. Design Alteration Warning: Use exact replacement parts only, i.e.,only those that are specified in the drawings and parts lists of this manual. This is especially important for the Interlock switches, described above. Never alter or add to the mechanical or electrical design of the MWO. Any design changes or additions will void the manufacturer's warranty. Always unplug the unit's AC power cord from the AC power source before attempting to remove or reinstall any component or assembly.
 14. Never defeat any of the B+ voltage interlocks. Do not apply AC power to the unit (or any of its assemblies) unless all solid-state heat sinks are correctly installed.
 15. Some semiconductor ("solid state") devices are easily damaged by static electricity. Such components are called Electrostatically Sensitive Devices (ESDs). Examples include integrated circuits and field-effect transistors. Immediately before handling any semiconductor components or assemblies, discharge the electrostatic charge from your body by touching a known earth ground.
 16. Always connect a test instrument's ground lead to the instrument chassis ground before connecting the positive lead; always remove the instrument's ground lead last.
 17. When checking the continuity of the switches or transformer, always make sure that the power is OFF, and one of the lead wires is disconnected.
 18. Components that are critical for safety are indicated in the circuit diagram by shading,  or .
 19. Use replacement components that have the same ratings, especially for flame resistance and dielectric strength specifications. A replacement part that does not have the same safety characteristics as the original might create shock, fire or other hazards.
- NOTE :** Connect the oven to a 20A circuit breaker. When connecting the oven to a 15A,make sure that circuit breaker can operate.

1. Precaution

1-2 Special High Voltage Precautions

1. High Voltage Warning Do not attempt to measure any of the high voltages --this includes the filament voltage of the magnetron. High voltage is present during any cook cycle. Before touching any components or wiring, always unplug the oven and discharge the high voltage capacitor (See Figure 1-1)
2. The high-voltage capacitor remains charged about 30 seconds after disconnection. Short the negative terminal of the high-voltage capacitor to to the oven chassis. (Use a screwdriver.)
3. High voltage is maintained within specified limits by close-tolerance, safety-related components and adjustments. If the high voltage exceeds the specified limits, check each of the special components.



PRECAUTION

There exists HIGH VOLTAGE ELECTRICITY with high current capabilities in the circuits of the HIGH VOLTAGE TRANSFORMER secondary and filament terminals. It is extremely dangerous to work on or near these circuits with the oven energized.

DO NOT measure the voltage in the high voltage circuit including filament voltage of magnetron.



PRECAUTION

Service technicians should remove their watches whenever working close to or replacing the magnetron.



PRECAUTION

Never touch any circuit wiring with your hand or with an uninsulated tool during operation.

2. Specifications

2-1 Features

- Professional Design

Samsung New OTR with Stainless Steel, it has professional look and the design matches great with other kitchen appliances, so that it provides harmonious kitchen interior and finally completes total kitchen solution.

- Turntable

Continually rotates food to ensure even cooking.

- Venting System

Four speed , 550CFM venting system

Quickly removes smoke and steam from cook top

- Ceramic coating

The interior fully coated with Ceramic Enamel offers solid surface that never scratches and cracks.

So you can enjoy the same new interior even after long use. Moreover, it prevents bacteria and the smooth surface is easy to clean as well.

The significantly higher cooking temperature and animal protein can cause severe discoloration and cleaning issues. But the Ceramic Enamel Interior prevents all of them.

- LED Cook-top Lighting

LED lighting gently brightens every corner of your cook top.

Low Power consumption and long life of Samsung LED light saves a remarkable amount of energy.

- LCD display

With an informative LCD, your OTR lets you easily use all available functions.

2. Specifications

2-2 Table of Specifications

Items			Basic Model	Derivated Model
			ME11A7510DS/AA ME11A7510DS/AC	ME11CB751012AA ME11CB751012AC
General	Capacity		1.1 Cu .Ft	1.1 Cu .Ft
	Power	Power Source	120V~60Hz	120V~60Hz
		Output Power (Microwave)	1,100W	1,100W
			1,000W	1,000W
		Power Consumption	1,650W	1,650W
			1,500W	1,500W
	Output Power		1100W/1000W (IEC-705)	1100W/1000W (IEC-705)
	Operation Frequescy		2,450MHz	2,450MHz
	Cooling Method		Cooling Fan Motor	Cooling Fan Motor
	Sensor		No	No
	Display		Ice Blue	White
	Contorl Method		Touch Glass	Touch Glass
	Door Open Mechanism		Side Swing	Side Swing
	Mw Distribution Mechanism		Side Feeding	Side Feeding
Feature	Sensor Cook Mode		No	No
	Auto Cook Mode		Yes	Yes
	Clock		Yes	Yes
	Timer		Yes	Yes
Feature	Max Time Cooking		99 Min 99 Sec	99 Min 99 Sec
	Power Level		Yes(10 Level)	Yes(10 Level)
	Setting	Sound Option	Yes	Yes
		Clock System Option (24h/12h/24h+12h)	24h + 12h	24h + 12h
		Child Lock	Yes	Yes
		Turntable On/Off	Yes	Yes
Material & Finishes	Door Color		Black Stss, Stss	WHITE
	Cavity Interior		Ceramic Enamel	Ceramic Enamel
	Packing Case		Yellow Box	Yellow Box
Accessory	Tray		Yes	Yes
	Cover Air (For Top Vent Option)		Yes	Yes

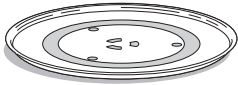
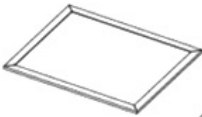
2. Specifications

2-2 Table of Specifications

Items		Basic Model	Derivated Model
		ME11A7510DS/AA ME11A7510DS/AC	ME11CB751012AA ME11CB751012AC
Physical Specification	Outside Dimensions	29 7/8"(W) X 10 3/4" (H) X 19 1/8" (D)	
	Oven Cavity Dimensions	22 1/8"(W) X 6 15/16" (H) X 14 27/32" (D)	
	Shipping Dimensions	33 1/4"(W) X 15 7/16" (H) X 22 1/8" (D)	
	Net Weight	23.5 Kg	24.3 Kg
	Shipping Weight	29.6 Kg	28.6 Kg
Export Zone		Usa Canada	Usa Canada

2. Specifications

2-3 Accessory

Item	Description	Code No.	Q'ty
	TRAY COOKING	DE63-00624A	1
	FILTER AIR	DE63-00944A	2
	FILTER CHARCOAL	DE63-00945A	2
	COVER AIR-LEFT	DE63-00946A	1
	COVER AIR-RIGHT	DE63-00947A	2
	ASSY GUIDE ROLLER	DE92-90495C	1

3. Disassembly and Reassembly

3-1 Tool for assembly and disassembly

MAGNETRON, MOTOR ASSEMBLY, VENT BLOWER AND HIGH VOLTAGE TRANSFORMER

Oven must be removed from wall.

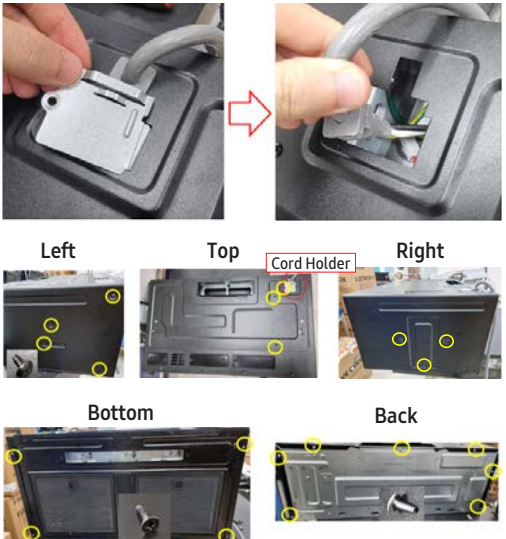
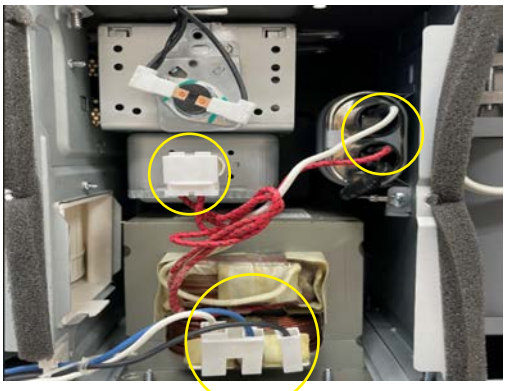
REMOVING OVEN FROM WALL (2 PEOPLE REQUIRED)

Oven is hooked on metal tabs at bottom of wall mounting plate and to cabinet by (2) top cabinet bolts.

Tool	Type	Remark
	Screw driver Star Screw driver	
	Nipper	

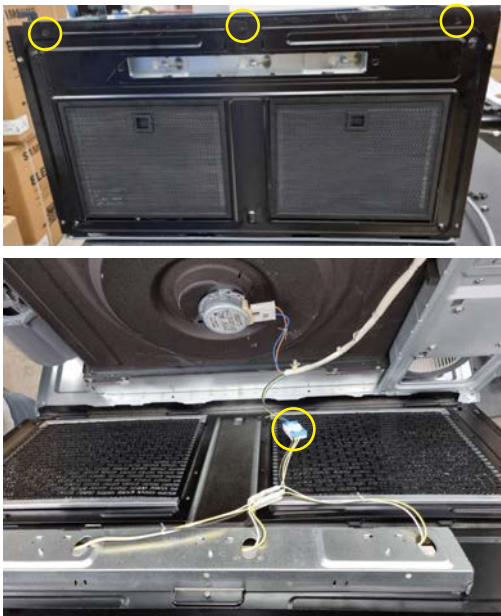
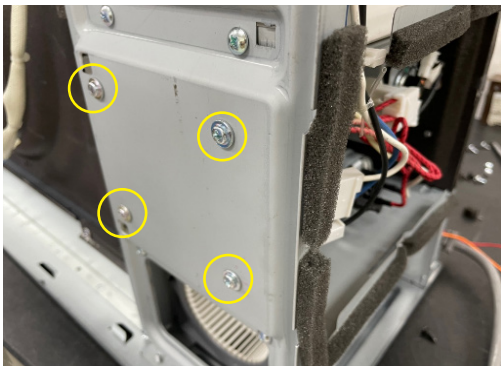
3. Disassembly and Reassembly

3-2 Replacement of High Voltage Transformer

Parts	Explanation Photo	Explanation
Panel Outer	 Left Top Cord Holder Right Bottom Back	1. Remove CORD HOLDER. 2. Remove PANEL SCREW at left, top, right, back side. 3. Rotate SET toward to up. 4. Remove PANEL OUTER screws at bottom and side. * Close or Disassemble ASSY DOOR while you remove screw at bottom side.
High Voltage Capacitor		1. Remove PANEL OUTER. 2. Discahrge the High voltage capacitor. 3. Disconnect all the leads.

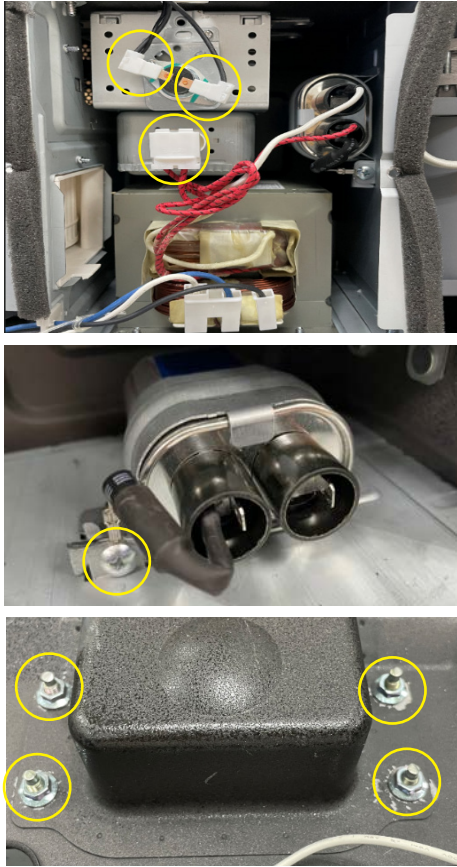
3. Disassembly and Reassembly

3-2 Replacement of High Voltage Transformer

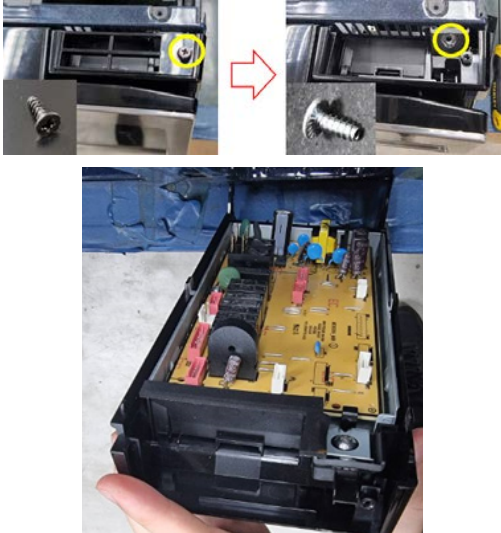
Parts	Explanation Photo	Explanation
Base Bottom		1. Remove PANEL OUTER. (12Page) 2. Remove 3 screws from the BASE BOTTOM. 3. Open the BASE BOTTOM. 4. Disconnect the wire connectors. Remove BASE BOTTOM.
HV Trans		5. Remove the 4 screws from the HV Transformer and take out the HV Trans.

3. Disassembly and Reassembly

3-3 Replacement of Magnetron

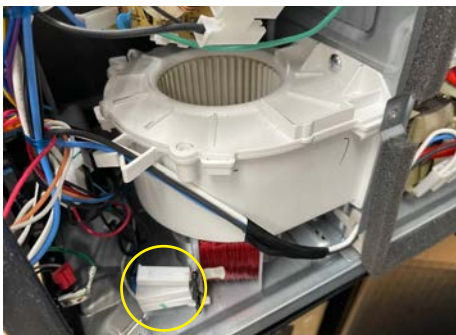
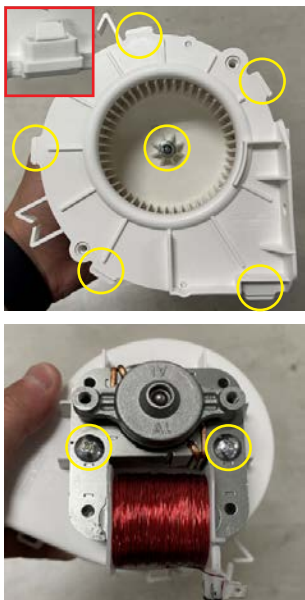
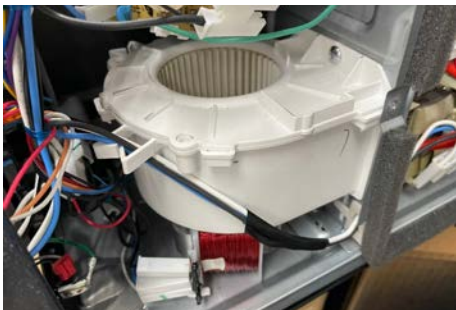
Parts	Explanation Photo	Explanation
Magnetron		Remove the magnetron including the shield case, permanent magnet, choke coils and capacitors (all of which are contained in one assembly). 1. Remove PANEL OUTER (12 Page) 2. Disconnect all lead wires from the magnetron. 3. Remove High Voltage Capacitor (12 Page) 4. Remove screw (1) securing C-OIL HIGH VOLTAGE. then also remove it. 5. Remove nuts (4) securing the magnetron to the wave guide. 6. Take out the magnetron very carefully.

3-4 Replacement of Fan Motor

Parts	Explanation Photo	Explanation
Fan Motor		1. Remove PANEL OUTER (12Page) and remove ASSY CONTROL BOX (20Page). 2. Disconnect the connectors from the control circuit board.

3. Disassembly and Reassembly

3-4 Replacement of Fan Motor

Parts	Explanation Photo	Explanation
Fan Motor		3. Disconnect all lead wires from FAN MOTOR.
		4. Remove BASE BOTTOM (page 13). 5. Turn the SET toward to upside. 6. Remove 2 screws securing ASSY MOTOR FAN.
		7. Disconnect 5 hooks than remove COVER BLOW-SUB. 8. Remove 1 Nut securing FAN MOTOR. 9. Remove 2 Screws securing COVER BLOWER.
		10. After replacing Fan Motor and reconnecting lead wires, make sure that the connectors are connected securely on the terminal of Fan Motor.

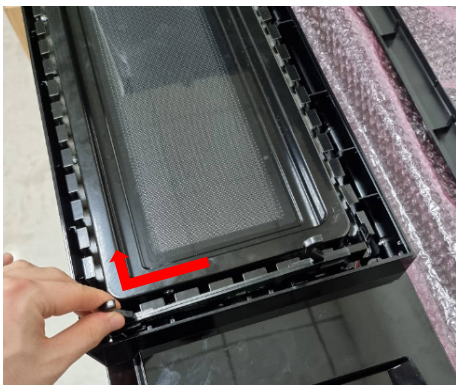
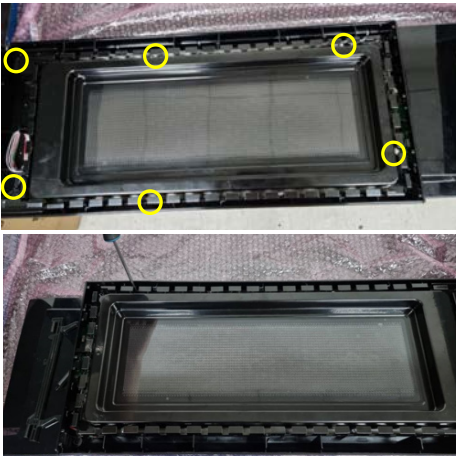
3. Disassembly and Reassembly

3-5 Replacement of Door Assembly

Parts	Explanation Photo	Explanation
Removal of Door Assembly		1. Disassemble CAP-DOOR 2. Disassemble 1screw and remove COVER AIR. 3. Diassemble COVER CONNECTOR and disconnect wire then disassemble 1screw. 4. Open the Assembly Door by degree 90° and remove COVER WIRE. ※ Pull out Cover Wire & Display wire at the same time. 5. Seperate lifting Assembly Door from Set.
Removal of Door C		5. Insert flat screwdriver into the gap between Door "A" and Door "C" to remove Door "C". • Be careful when handling Door "C" because it is fragile.

3. Disassembly and Reassembly

3-5 Replacement of Door Assembly

Parts	Explanation Photo	Explanation
Removal of Key Door & Spring		6. Detach the spring and key door from Door.
Removal of Door "e"		7. Remove 6 screws, securing Door 'E' and detach from Door "A". 8. Following the procedure as shown in the figure, insert Flat driver between Door "E" and Door "A" * Be careful when handling Door "E" because it affects microwave leakage.

3. Disassembly and Reassembly

3-6 Replacement of Assy Noise Filter

Parts	Explanation Photo	Explanation
ASSY NOISE FILTER		1. Disconnect Power. 2. Remove CONTROL-BOX (19Page) 3. Remove ground screw. 4. Replace ASSY NOISE FILTER 5. Assemble ground.

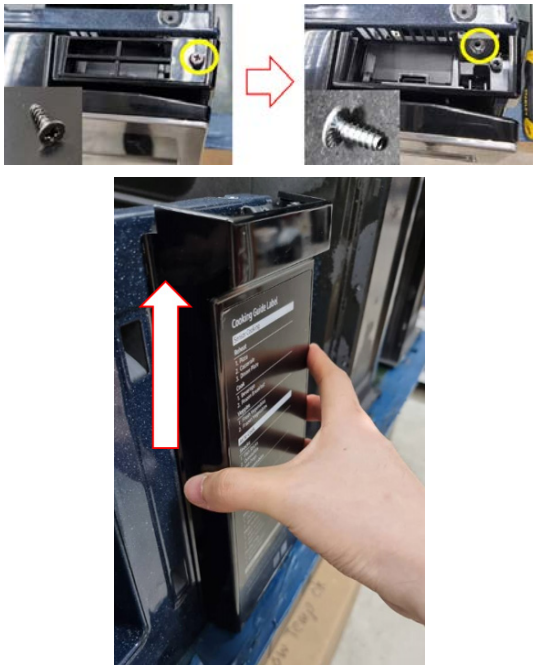
3-7 Replacement of Drive Motor

Parts	Explanation Photo	Explanation
Drive Motor		1. Remove BASE BOTTOM (Page13) 2. Disconnect power and remove bottom plate 3 screws. 3. Remove bottom plate and disconnect the turntable motor drive. 4. Remove turntable motor screw(1) and pull the turntable motor out. 5. When replacing the drive motor, be sure to remount it in the correct position with the coupler. 6. Connect all the leads to the drive motor.

3. Disassembly and Reassembly

3-8 Replacement of Control Circuit Board

3-8-1 Main PBA

Parts	Explanation Photo	Explanation
Removal of Control Box Assembly		1. Disconnect power. 2. Be sure to ground any static electric charge in your body and never touch the control circuit. 3. Remove a screw securing the control box vent assembly. 4. Remove a screw securing the control box and cavity. 5. Slide the control box upward and pull out the control box.
Removal of P.C.B Assembly		1. Disconnect wire from circuit board * Pull lightly 2. Remove a screw securing the circuit board. 3. Lift up the control circuit board from the Ass'y control box. 4. When reconnecting the FPC connector, make sure that the holes on the connector are properly engaged with the hooks on the Plastic Fastener. 5. After reconnecting the connectors, make sure that the connectors are connected securely on the terminal of control circuit board by pulling the wires lightly.

3. Disassembly and Reassembly

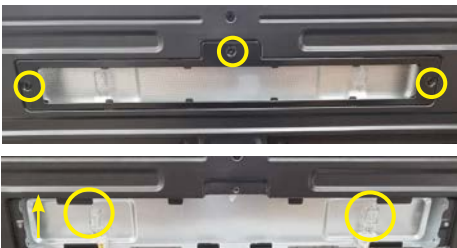
3-8 Replacement of Control Circuit Board

3-8-2 Sub Module

Parts	Explanation Photo	Explanation
Removal of PCB Assembly (Sub Module)		1. Remove DOOR "E" (Page 17) 2. Remove 2screws. 3. Insert Flat driver between Door'A' and Cover handle 4. Open up Cover handle and remove it. 5. Remove 2screws of Door Circuit. 6. Remove Wire connect from Circuit. 7. Lift up the circuit from the Door 'A'

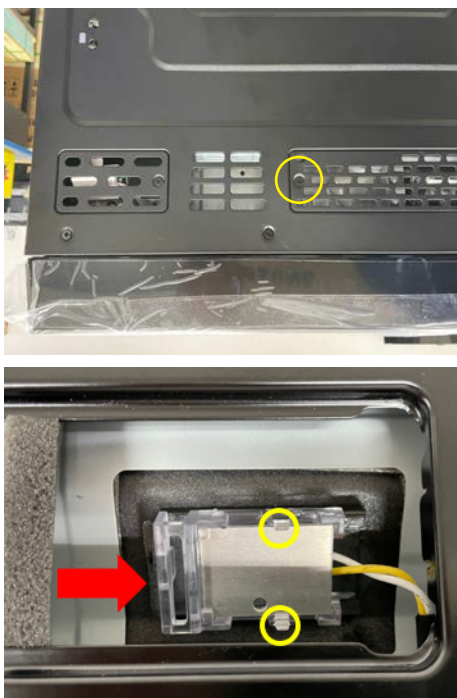
3. Disassembly and Reassembly

3-9 Replacement of Cooktop Lamp

Parts	Explanation Photo	Explanation
Cooktop Lamp(LED)		1. Unplug the oven or turn off the power at the main power. 2. Remove 3 screw on cover lamp. 3. Unplug the LED wire and remove screw. 4. Replace the LED and plug wire and install screw. 5. Turn the power back on at the main power supply.
Cooktop Lamp (HALOGEN LAMP)		1. Unplug the oven or turn off the power at the main power. 2. Remove 3 screw on cover lamp. 3. Unplug the HALOGEN LAMP. 4. Replace the HALOGEN LAMP and install screw. 5. Turn the power back on at the main power supply.

3. Disassembly and Reassembly

3-10 Replacement of Oven Light

Parts	Explanation Photo	Explanation
Oven Light		1. Remove screw (1) securing with COVER AIR. 2. Remove HOLDER LAMP lift and remove the holder - lamp while pressing the hook. 3. Disconnect wire-harness. 4. Pull the hook on the LED lamp and remove it. 5. Replace the new LED lamp.

4. Alignment and Adjustments



PRECAUTION

1. High voltage is present at the high voltage terminals during any cook cycle.
2. It is neither necessary nor advisable to attempt measurement of the high voltage.
3. Before touching any oven components or wiring, always unplug the oven from its power source and discharge the high voltage capacitor.

4-1 High Voltage Transformer

1. Remove connectors from the transformer terminals and check continuity.
2. Normal resistance readings are as follows:

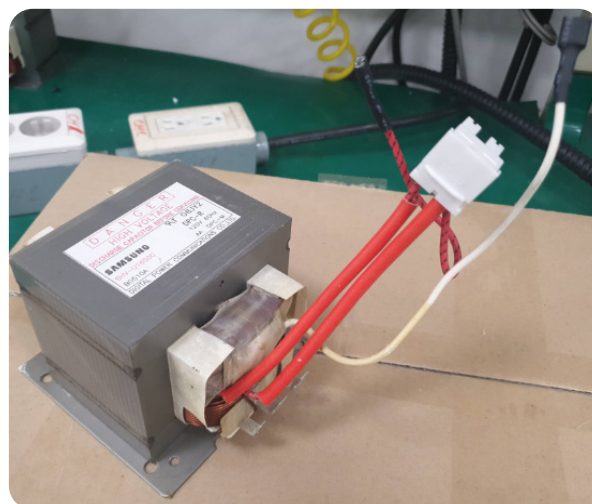
◁ ME11A7710*/AA, ME11A7510*/AA ▷

	SHV-U1870E
Secondary	Approx. 123.5 Ω + 2%
Filament	Shows Continuity
Primary	Approx. 0.410 Ω + 2% (High) Approx. 0.475 Ω + 2% (Low)

◁ ME11A7710*/AC, ME11A7510*/AC ▷

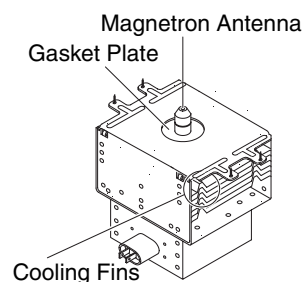
	SHV-U1850C
Secondary	Approx. 131.0 Ω ± 2%
Filament	Shows Continuity
Primary	Approx. 0.440 Ω + 2% (High) Approx. 0.480 Ω + 2% (Low)

(Room temperature = 20°C)



4-2 Magnetron

1. Continuity checks can indicate only an open filament or a shorted magnetron. To diagnose an open filament or shorted magnetron.
2. Isolate the magnetron from the circuit by disconnecting its leads.
3. A continuity check across the magnetron filament terminals should indicate one ohm or less.
4. A continuity check between each filament terminal and magnetron case should read open.



4-3 High Voltage Capacitor

1. Check continuity of the capacitor with the meter set at the highest resistance scale.
2. Once the capacitor is charged, a normal capacitor shows continuity for a short time, and then indicates 9M Ω .
3. A shorted capacitor will show continuous continuity.
4. An open capacitor will show constant 9M Ω .
5. Resistance between each terminal and chassis should read infinite.

4. Alignment and Adjustments

4-4 High Voltage Diode

1. Isolate the diode from the circuit by disconnecting its leads.
2. With the ohm-meter set at the highest resistance scale, measure across the diode terminals. Reverse the meter leads and read the resistance. A meter with 6V, 9V or higher voltage batteries should be used to check the front-to-back resistance of the diode (otherwise an infinite resistance may be read in both directions). The resistance of a normal diode will be infinite in one direction and several hundred K Ω in the other direction.

4-5 Main Relay and Power Control Relay

1. The relays are located on the PCB Ass'y. Isolate them from the main circuit by disconnecting the leads.
2. Operate the microwave oven with a water load in the oven. Set the power level set to high.
3. Check continuity between terminals of the relays after the start pad is pressed.

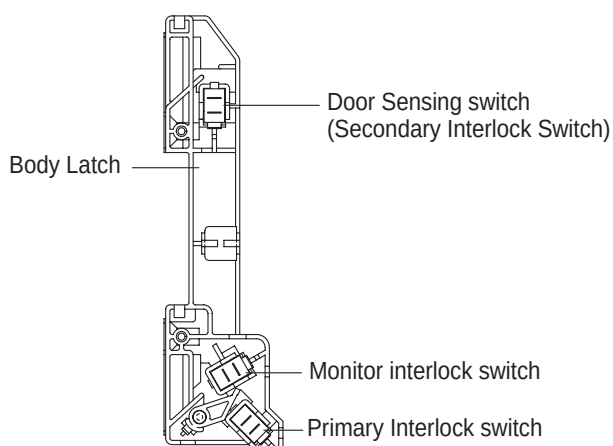
4-6 Adjustment of Primary interlock switch, Door sensing switch (Secondary interlock switch), and Monitor interlock switch



PRECAUTION

For continued protection against radiation hazard, replace parts in accordance with the wiring diagram and be sure to use the correct part number for the following switches: Primary and secondary interlock switches, and the monitor interlock switch (replace all together). Then follow the adjustment procedures below. After repair and adjustment, be sure to check the continuity of all interlock switches and the monitor interlock switch.

1. When mounting Primary interlock switch and Monitor interlock switch to Latch Body, consult the figure.
2. No specific adjustment during installation of Primary interlock switch and Monitor interlock switch to the latch body is necessary.
3. When mounting the Latch Body to the oven assembly, adjust the Latch Body by moving it so that the oven door will not have any play in it. Check for play in the door by pulling the door assembly. Make sure that the latch keys move smoothly after adjustment is completed. Completely tighten the screws holding the Latch Body to the oven assembly.
4. Reconnect to Monitor switch and check the continuity of the monitor circuit and all latch switches again by following the components test procedures.
5. Confirm that the gap between the switch housing and the switch actuator is no more than 0.5mm when door is closed.
6. Interlock Switch Replacement - When replacing faulty switches, be sure switch mounting tabs are not bent, broken or otherwise deficient in their ability to secure the switches in place.



	Door Open	Door Closed
Primary Interlock switch	∞	0
Monitor interlock switch(COM-NC)	0	∞
Monitor interlock switch(COM-NO)	∞	0
Door Sensing switch (Secondary Interlock switch)	∞	0

4. Alignment and Adjustments

4-7 Vent Exhaust Blower Motor

THIS COMPONENT REQUIRES REMOVAL OF MICROWAVE OVEN FROM INSTALLATION FOR SERVICING.

RUN CAPACITOR

The run capacitor is located behind the mgt. The capacitor is used for more torque and electrical phasing. Without the capacitor the blower might run but would be much slower.

TO TEST THE CAPACITOR

1. Discharge capacitor and disconnect one capacitor lead.
2. Make appropriate capacitor check (with analog meter needle should rise then fall, cap is charging then discharging.)

TO TEST BLOWER WINDINGS:

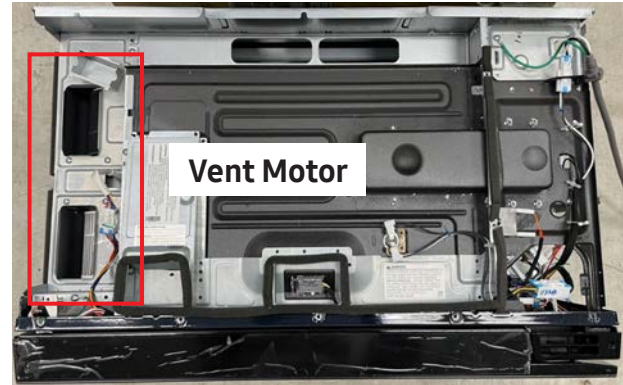
1. Disconnect power.
2. Open PANEL OUTER and discharge capacitor
3. Disconnect 2 wires to run capacitor. Continuity test across the 2 wires should be approximately 94 ohms of resistance. This test allows you to read across all 3 windings at the same time.

TO REMOVE VENT BLOWER

1. Remove unit from its installation.
2. Remove PANEL OUTER
3. Disconnect blower plug.
4. Remove 4 screws securing bracket motor. Pull it forward and pull it out.

NOTE: Place blower wires in routing slots to avoid pinching of wires.

NOTE: It's normal that connector bracket bends when reconnecting Hood Blower Motor connector.



4. Alignment and Adjustments

4-8 Thermal Cutout (TCO'S)

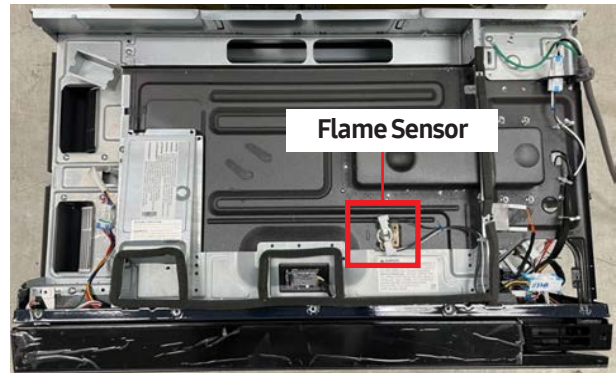
There are 4 different thermal cutouts in this unit with 4 different purposes. They are :

1. Oven thermal cutout(Flame sensor), on cavity top.
2. Hood thermal cutout, on floor of control area.
3. Bottom thermal cutout, on floor of control area.
4. Magnetron thermal cutout, on magnetron.

4-8-1 Oven Thermal Cutout(Flame sensor)

The oven thermal cutout(Cavity TCO) is located on the top side of the oven cavity with a temperature rating of 248°F(120°C) and is non-resettable.

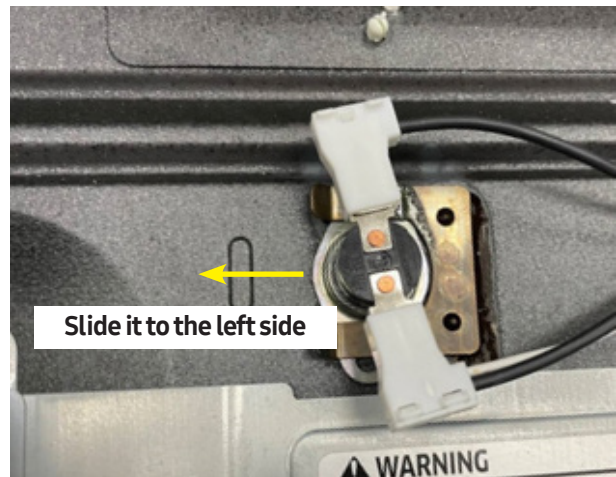
The cutout is tightly held to the top of the oven cavity.



4-8-2 Replacement of Flame Sensor

Disconnect oven power. Remove PANEL OUTER

1. Disconnect 2 wire leads.
2. Slide the top of the thermal cutout and pull out it.



4. Alignment and Adjustments

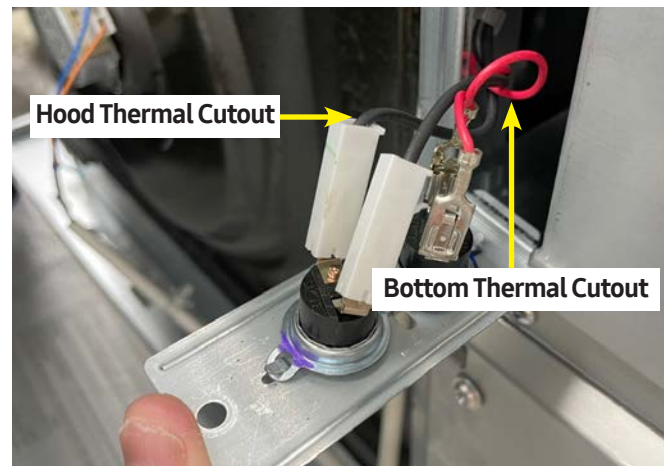
4-8-3 Hood Thermal Cutout

This cutout will protect the touch control from excessive heat by turning the vent fan on at low speed. If the surface units of the range are used for long periods of time heat will build up and could damage the microwave control. In order to prevent this a thermal cutout is installed on the duct behind the control. This cutout will close (158°F/70°C - vent fan energized) and open (104°F/40°C - vent fan de-energized) depending on temperatures it senses.

To Remove hood Thermal Cutout :

1. Disconnect power .
2. Remove BASE BOTTOM.
3. Remove screw (1) securing BRACKET ELEC
4. Pull out BRACKET ELEC.

NOTE : If this cutout were to open it would be difficult to detect. The only time it functions is during an overheat condition. It will be normally open when checked with an ohmmeter.



4-8-4 Bottom Thermal Cutout

This part can be serviced without removing the microwave from installation.

During a fire on the stove the heat could be intense enough to close the Hood Thermal Cutout and force the fan to run. While at moderate high temperature we do want it to run, during a fire it is advantageous to NOT have the vent fan running. So if a fire were to start on the stove top the Bottom Thermal Cutout would open at 248°F(120°C) and remove all power to the microwave oven.

This cutout is designed to not be resettable.

To Remove Magnetron Thermal Cutout :

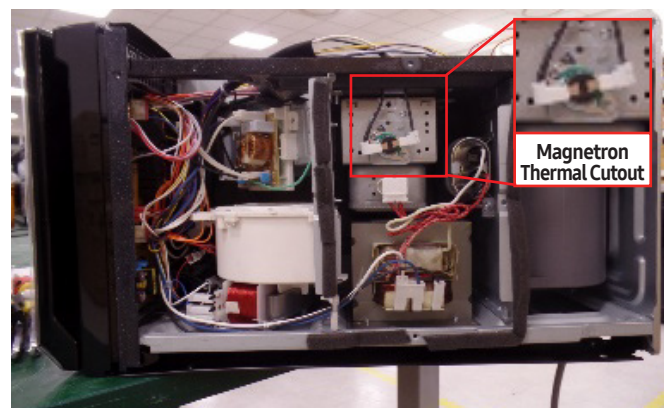
1. Disconnect power.
2. Remove PANEL-OUTER.
3. Remove 2 wire leads and remove screw (1) securing the cutout on the MGT

4-8-5 Magnetron Thermal Cutout

The magnetron thermal cutout is located above the leads to the magnetron. It is designed to prevent damage to the magnetron if an overheated condition develops in the tube due to a cooling fan failure, obstructed air ducts, or a dirty or blocked air intake. Under normal operation, the magnetron thermal cutout remains closed. However, when abnormally high temperatures are reached within the magnetron, the magnetron thermal cutout will open at 302°F(150°C) causing the oven to shut down. After the temperature drops to 140° F(60°C) it will reset and cooking will be able to resume.

To Remove Magnetron Thermal Cutout :

1. See 'Removing Magnetron'.



4. Alignment and Adjustments

4-9 Output Power of Magnetron



PRECAUTION

MICROWAVE RADIATION

PERSONNEL SHOULD NOT ALLOW EXPOSURE TO MICROWAVE RADIATION FROM MICROWAVE GENERATOR OR OTHER PARTS CONDUCTING MICROWAVE ENERGY.

The output power of the magnetron can be measured by performing a water temperature rise test.

Equipment needed :

- Two 1-liter cylindrical borosilicate glass vessel (Outside diameter 190 mm)
 - One glass thermometer with mercury column
- NOTE: Check line voltage under load. Low voltage will lower the magnetron output. Make all temperature and time tests with accurate equipment.

1. Fill the one liter glass vessel with water.
2. Stir water in glass vessel with thermometer, and record glass vessel's temperature ("T1", 10±1°C).
3. After moving the water into another glass vessel, place it in the center of the cooking tray. Set the oven to high power and operate for 45 seconds exactly. (3 seconds included as a holding time of magnetron oscillation:)
4. When heating is finished, stir the water again with the thermometer and measure the temperature ("T2").
5. Subtract T1 from T2. This will give you the water temperature rise. (ΔT)
6. The output power is obtained by the following formula;

$$\text{Output Power} = \frac{4.187 \times 1000 \times \Delta T + 0.55 \times M_c \times (T_2 - T_1)}{42}$$

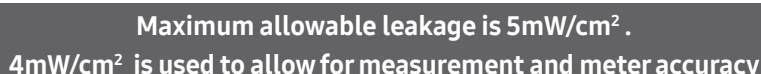
45	: Heating Time
42	: Counting Time (sec)
4.187	: Coefficient for Water
1000	: Water (cc)
ΔT	: Temperature Rise (T2-T1)
To	: Room Temperature
Mc	: Cylindrical borosilicate glass weight

7. Normal temperature rise for this model is 9°C to 11°C at 'HIGH'.

NOTE 1: Variations or errors in the test procedure will cause a variance in the temperature rise. Additional power test should be made if temperature rise is marginal.

NOTE 2: Output power in watts is computed by multiplying the temperature rise (step 5) by a factor of 91 times the of centigrade temperature.

1. Pour 275±15cc of 20±5°C(68±9°F) water in a beaker which is graduated to 600cc, and place the beaker in the center of the oven.
2. Start to operate the oven and measure the leakage by using a microwave energy survey meter.
3. Set survey meter with dual ranges to 2,450MHz.
4. When measuring the leakage, always use the 2 inch spacer cone with the probe. Hold the probe perpendicular to the cabinet door. Place the spacer cone of the probe on the door and/or cabinet door seam and move along the seam, the door viewing window and the exhaust openings moving the probe in a clockwise direction at a rate of 1 inch/sec. If the leakage testing of the cabinet door seam is taken near a corner of the door, keep the probe perpendicular to the areas making sure that the probe end at the base of the cone does not get closer than 5cm to any metal. If it gets closer than 5cm, erroneous readings may result.
5. Measured leakage must be less than 4mW/cm² , after repair or adjustment.



1. Do not exceed the limited scale.
2. The test probe must be held on the grip of the handle, otherwise a false reading may result when the operator's hand is between the handle and the probe.
3. When high leakage is suspected, do not move the probe horizontally along the oven surface; this may cause damage to the probe.
4. Follow the recommendation of the manufacturer of the microwave energy survey meter.

4-13-2 At least once a year have the microwave energy survey meter checked for accuracy by its manufacturer.

4. Alignment and Adjustments

4-14 Sensor

The Sensor Cooking Function uses a special gas sensor which detects both humidity(steam) and hydrocarbons(food odors) during the cooking process. Before conducting either of the sensor tests below, ensure the unit is plugged into a wall outlet for at least 5 minutes. If already plugged in, proceed. The sensor is a plug-in device located in the vent area at the top left hand corner of the cavity behind the grill.

To Service :

1. Disconnect power and remove PANEL-OUTER.
2. Separate ASSY DUCT-UP and separate retainer from receptacle and unplug sensor.

SENSOR TEST (QUICK TEST)

1. With 2 fingers touch and hold the following pads at the same time by standby mode(Display Clock condition):

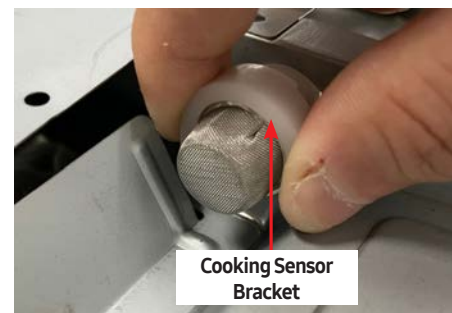
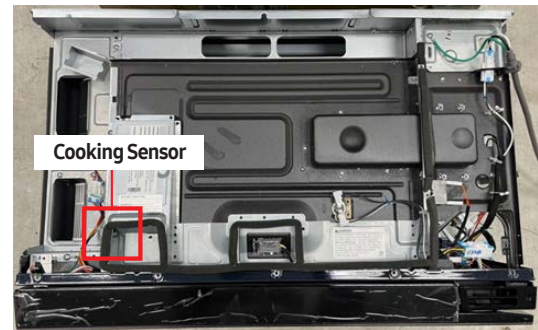
Auto Defrost

 and

Popcorn
2. Observe diagnostic number in display (Numbers approximate).
 15-185
 (Normal-verify with "detection test")
 213 or Higher
 (Sensor failed to open, sensor unplugged, wiring or smart board.)
 Less then 6
 (shorted sensor, or smart board).

NOTE : Only heater terminals (H ; Black and Red leads) can be checked with ohmmeter(30Ω).

CAUTION : DO NOT ATTEMPT TO CHECK SENSOR TERMINALS (White and Orange leads).
CAN DAMAGE SENSOR.



5. Troubleshooting

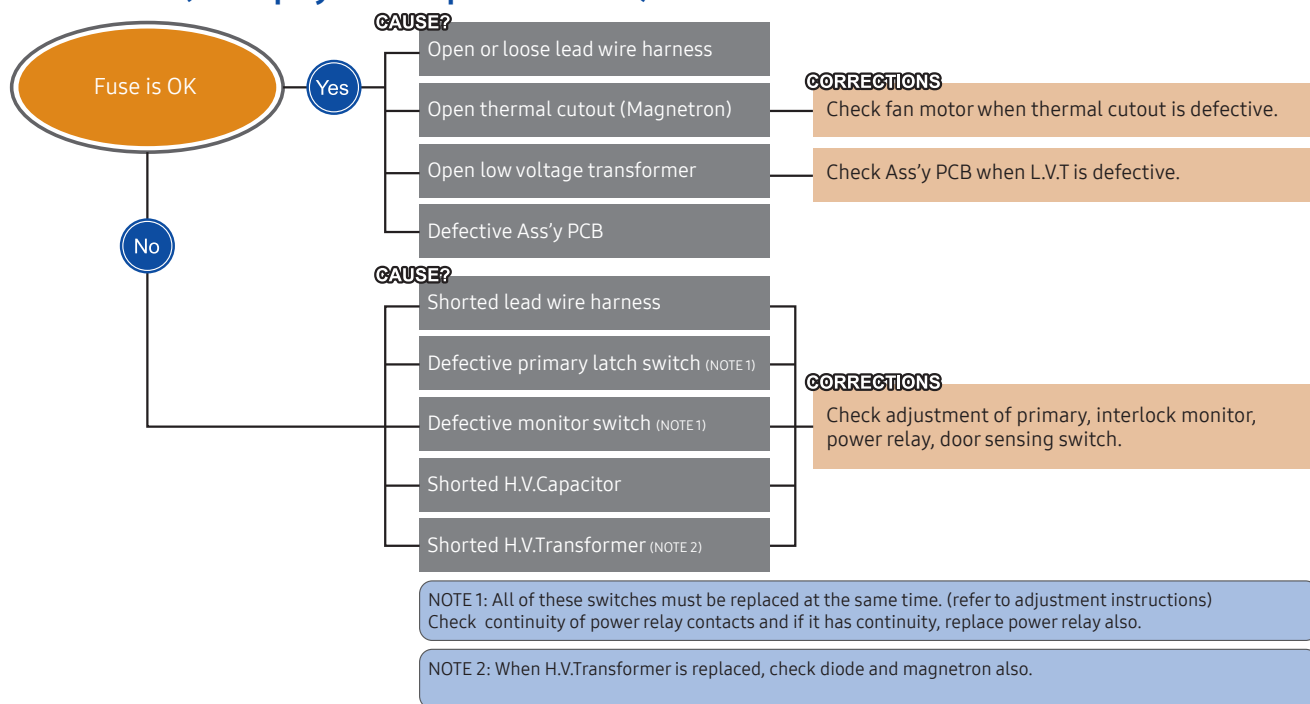


PRECAUTION

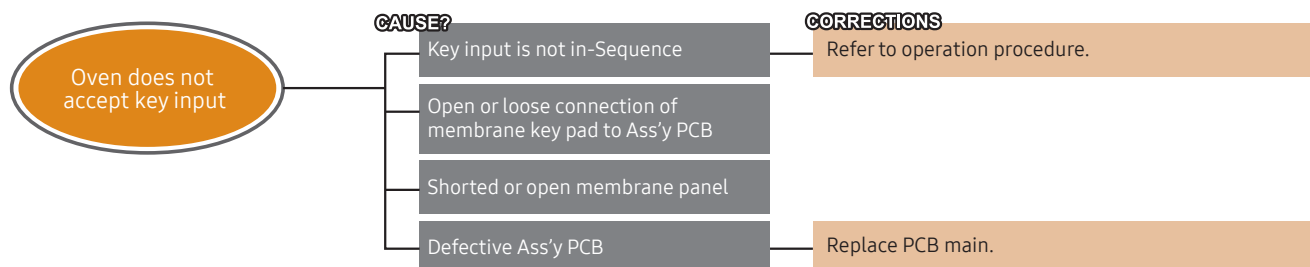
1. CHECK GROUNDING BEFORE CHECKING FOR TROUBLE.
2. BE CAREFUL OF THE HIGH VOLTAGE CIRCUIT.
3. DISCHARGE THE HIGH VOLTAGE CAPACITOR.
4. WHEN CHECKING THE CONTINUITY OF THE SWITCHES OR TRANSFORMER, DISCONNECT ONE LEAD WIRE FROM THESE PARTS AND THEN CHECK CONTINUITY WITHOUT THE POWER SOURCE ON. TO DO OTHERWISE MAY RESULT IN A FALSE READING OR DAMAGE TO YOUR METER.
5. DO NOT TOUCH ANY PART OF THE CIRCUIT OR THE CONTROL CIRCUIT BOARD, SINCE STATIC DISCHARGE MAY DAMAGE IT. ALWAYS TOUCH GROUND WHILE WORKING ON IT TO DISCHARGE ANY STATIC CHARGE BUILT UP.

5-1 Electrical Malfunction

Oven is dead. (No display and no operation at all.)



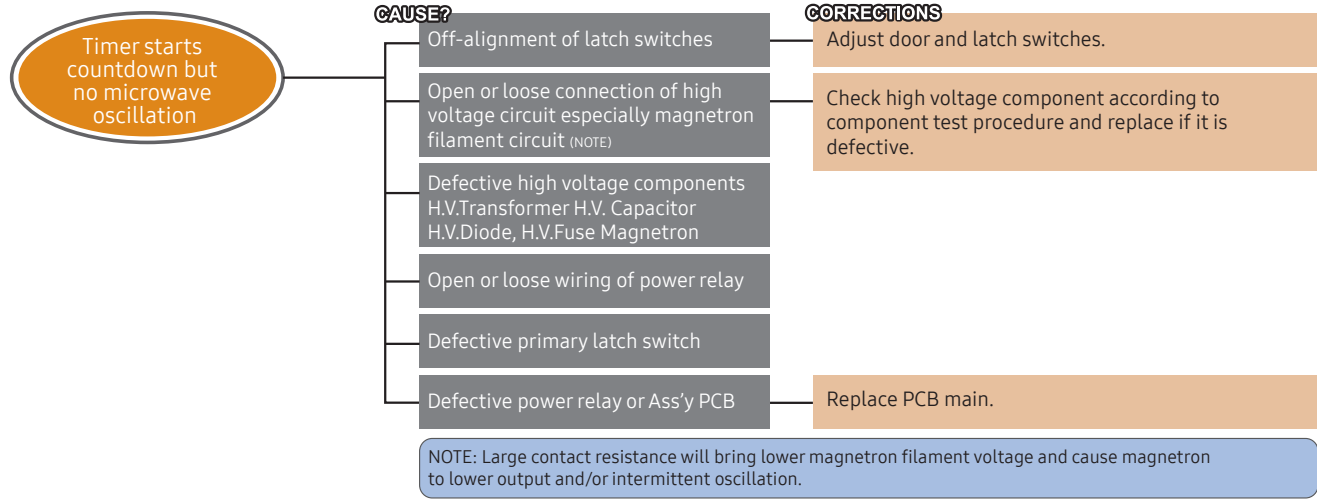
Oven does not accept key input (Program)



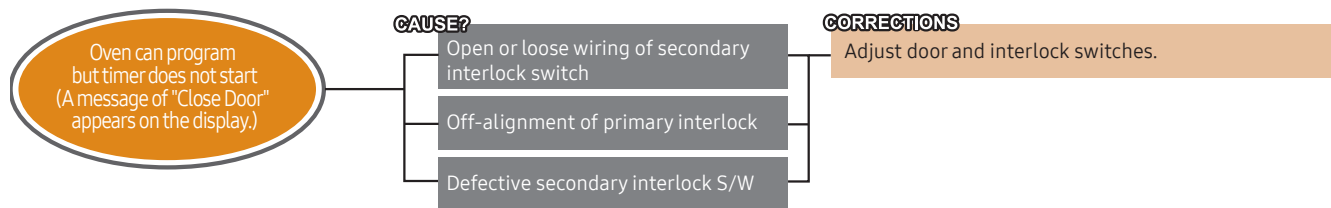
5. Troubleshooting

Timer starts countdown but no microwave oscillation.

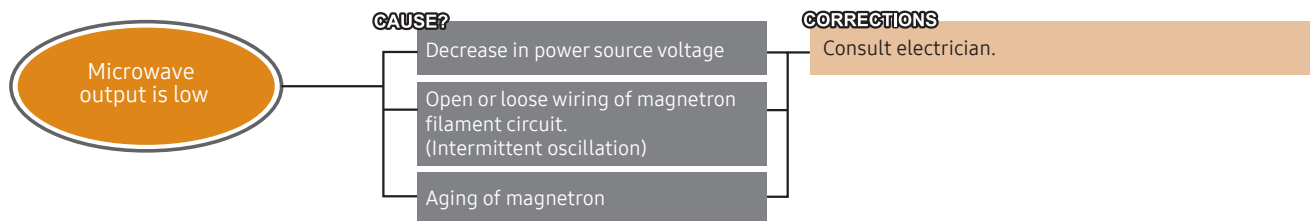
(No heat while oven lamp and fan motor turn on.)



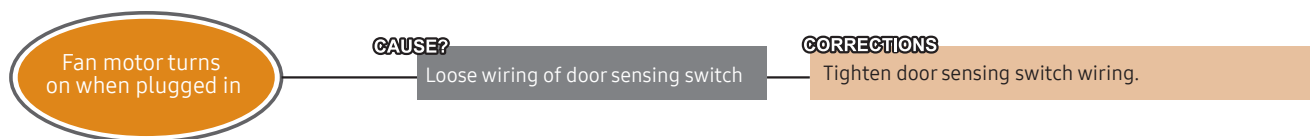
Oven can program but timer does not start.(A message of "Close Door" appears on the display.)



Microwave output is low. (Oven takes longer time to cook food.)



Fan motor turns on when plugged in.

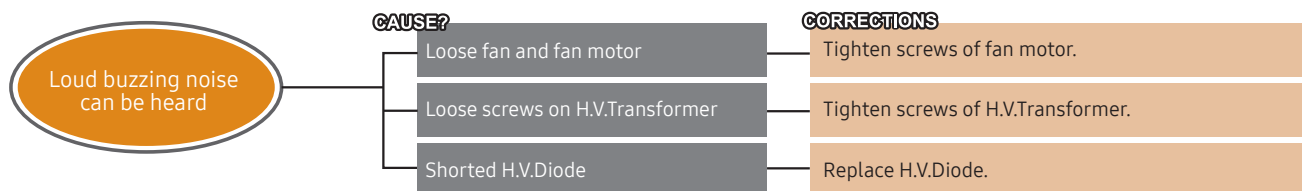


Oven does not operate and return to the plugged in mode.

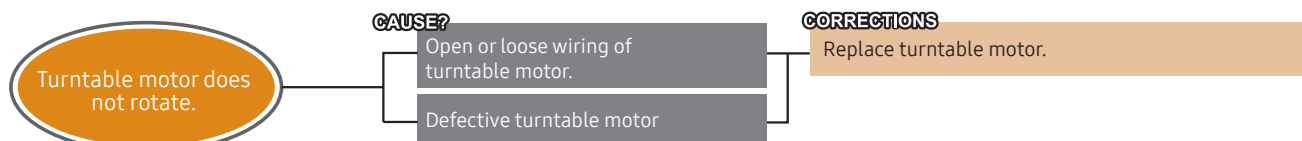


5. Troubleshooting

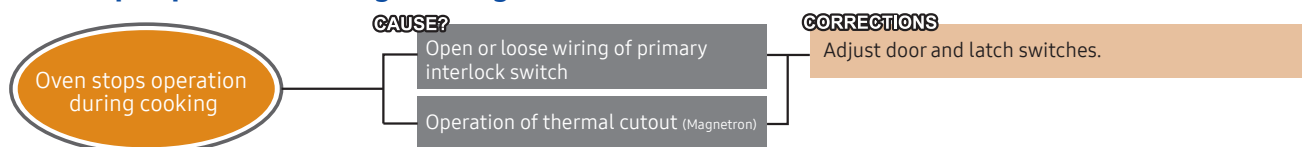
Loud buzzing noise can be heard.



Turntable motor does not rotate.



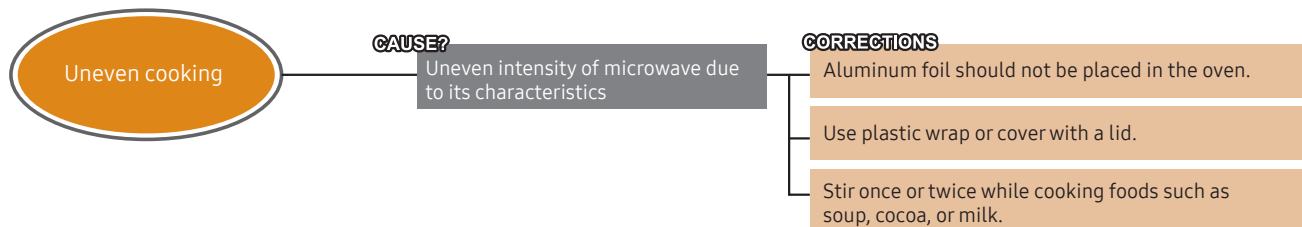
Oven stops operation during cooking.



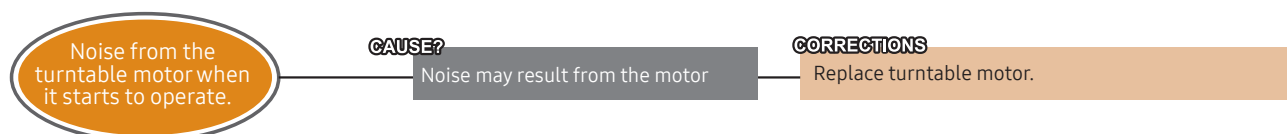
Sparks.



Uneven cooking.



Noise from the turntable motor when it starts to operate.



5. Troubleshooting

5-2 Information Code Numbering Rule

1. Information CODE NUMBERING RULE is applied to a microwave oven and an oven.
(CMO, OTR, Grill, Convection, Commercial etc.)
2. All sensors and devices have their own number.
ex) Gas/Easy/PH Sensor = 1, Temp. Sensor = 2, ...
3. Of each device, No.0~2 refer to Information Option 1~3



0 - Others

1 – Gas/Easy/PH Sensor

2 – Temp Sensor

7 – Steam Sensor

A – Function

d – Door/Key

F – Main/Sub Communication

X – additional Information

0 - Information option 1

1 – Information option 2

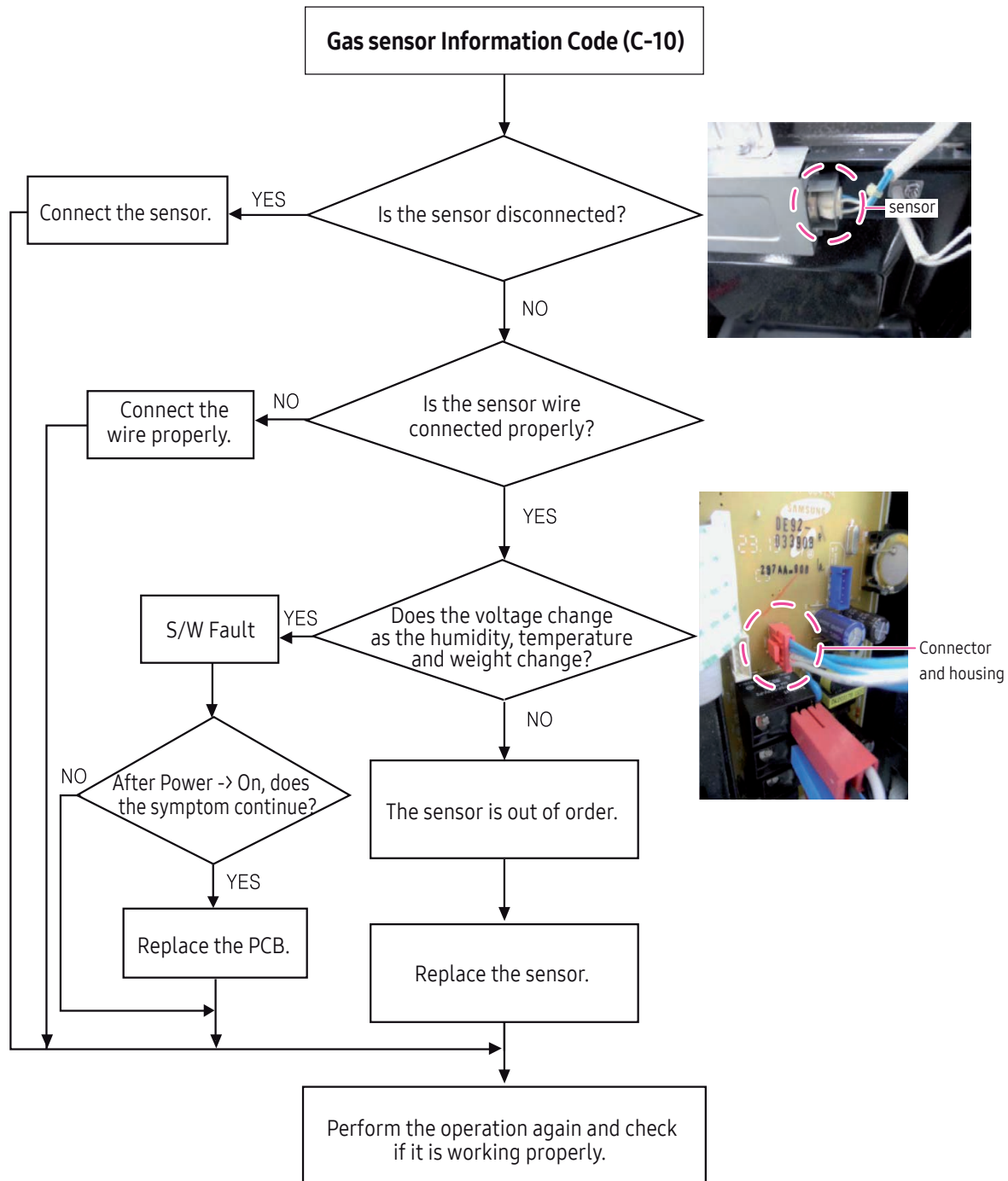
2 – Information option 3

5-3 Information Code List

Code	Symptom	Solution
C-20	Temp Sensor Error	Check the sensor unit and point the connection between the sensor terminal and the substrate. The temperature sensor short circuit is black.
C-21	When detecting abnormal temperature in microcooking	Cooling down the unit for awhile and re-operate
C-F0	PCB Error	Check the connector between the main PCB and the sub PCB.
C-d0	Key Short	Replace buttons and touch parts or restart
C-F1	PBA defect	PBA Change
C-F2	Touch defect	Check the Touch Film Tale and Connector

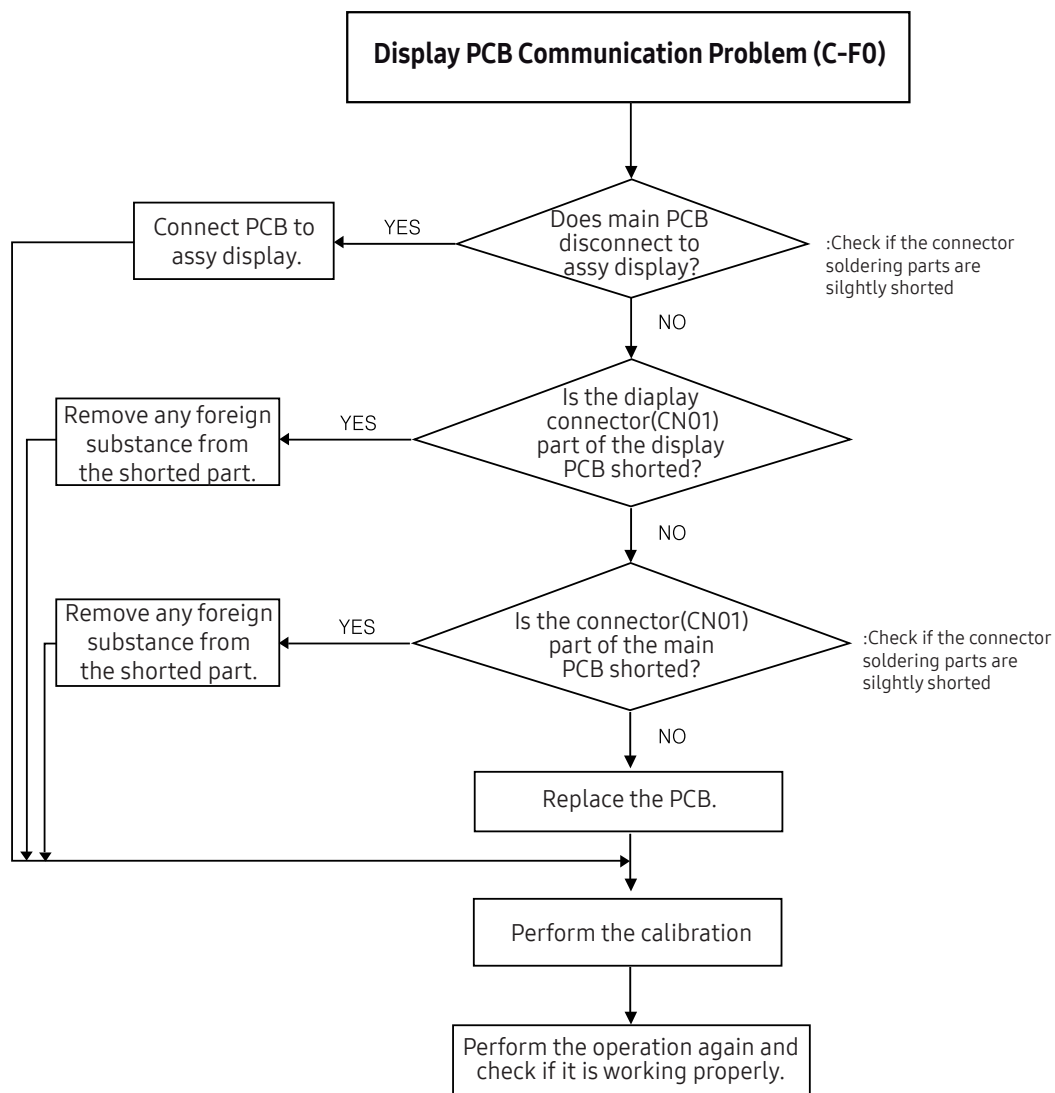
5. Troubleshooting

5-3-1 C-10



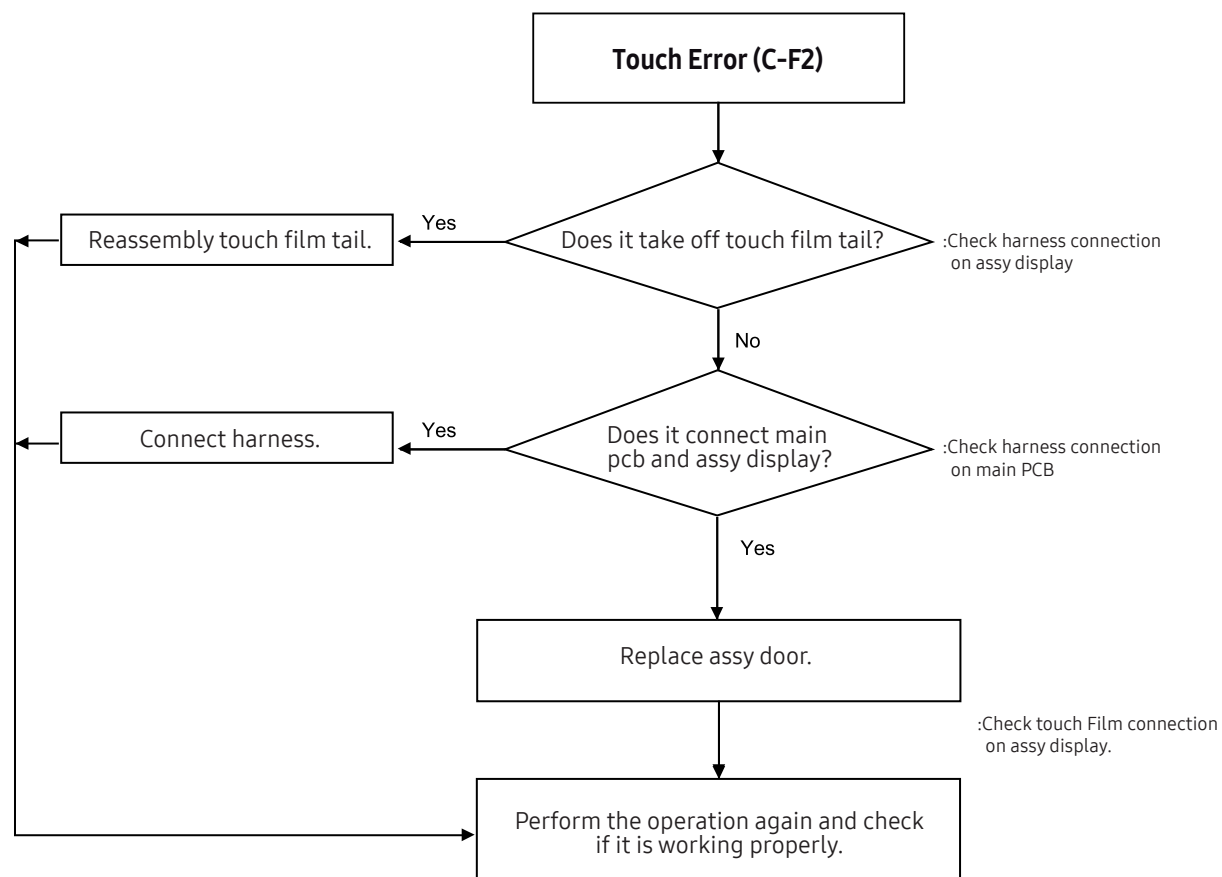
5. Troubleshooting

5-3-2 C-F0



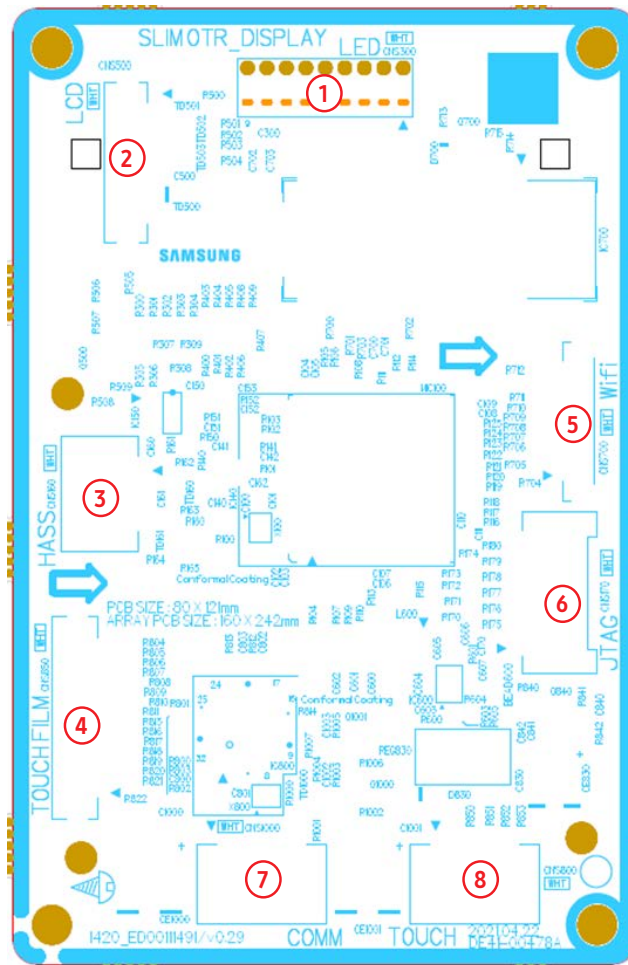
5. Troubleshooting

5-3-3 C-F2



6. PCB Diagrams

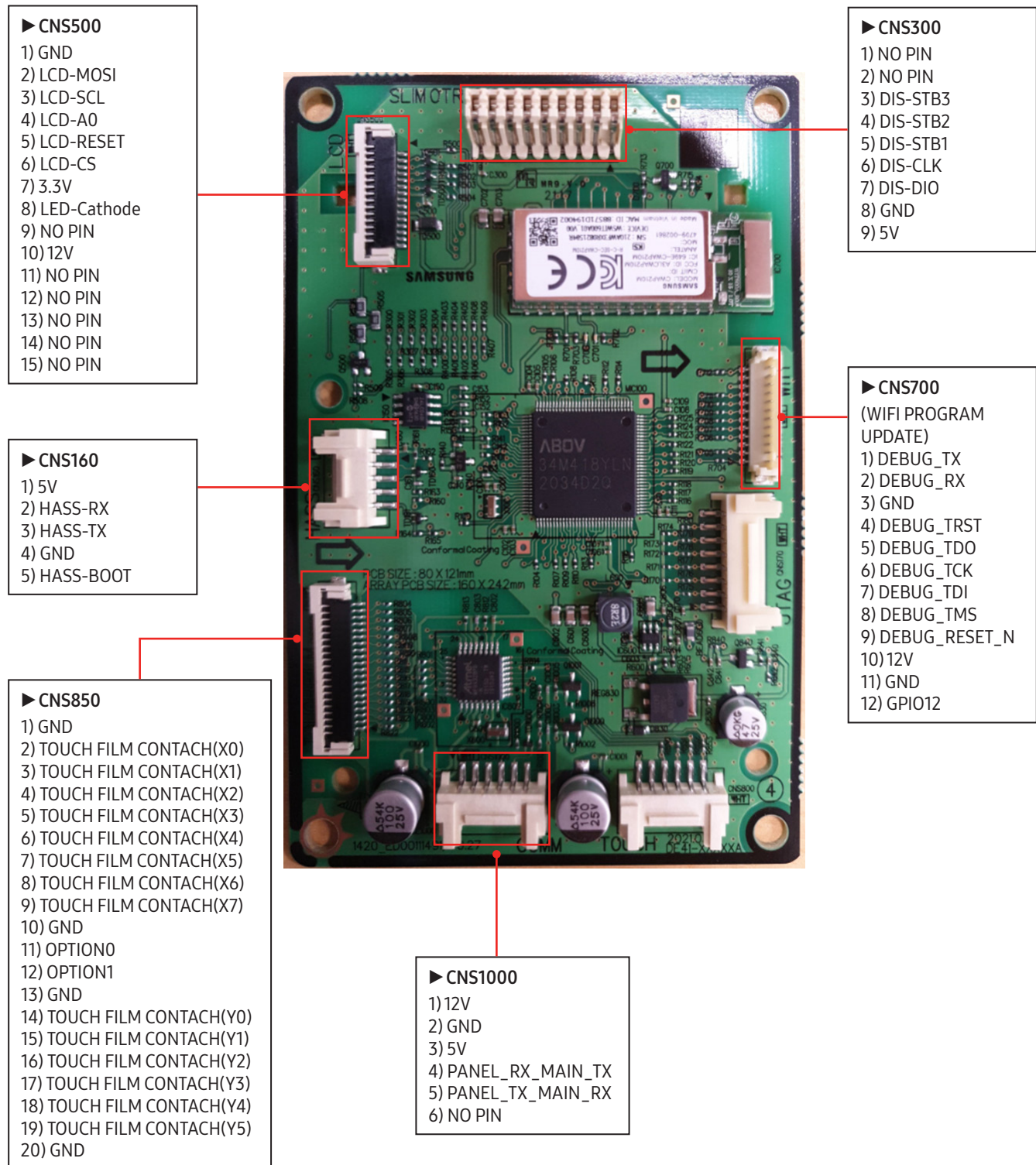
6-1 PCB Diagrams (Sub Module)



No.	Part Number	Part Name	Function and Rule
1	CNS300	LED Display Connector	LED Display
2	CNS500	MONO LCD Connector	MONO LCD
3	CNS160	DOWNLOAD & HASS Connector	DOWNLOAD & HASS
4	CNS850	TOUCH FILM Connector	TOUCH FILM
5	CNS700	WIFI DOWNLOAD Connector	WIFI DOWNLOAD
6	CNS170	JTAG Connector	JTAG
7	CNS1000	SUB-MAIN Communication Connector	SUB-MAIN Communication
8	CNS800	Touch DOWNLOAD Connector	TOUCH DOWNLOAD

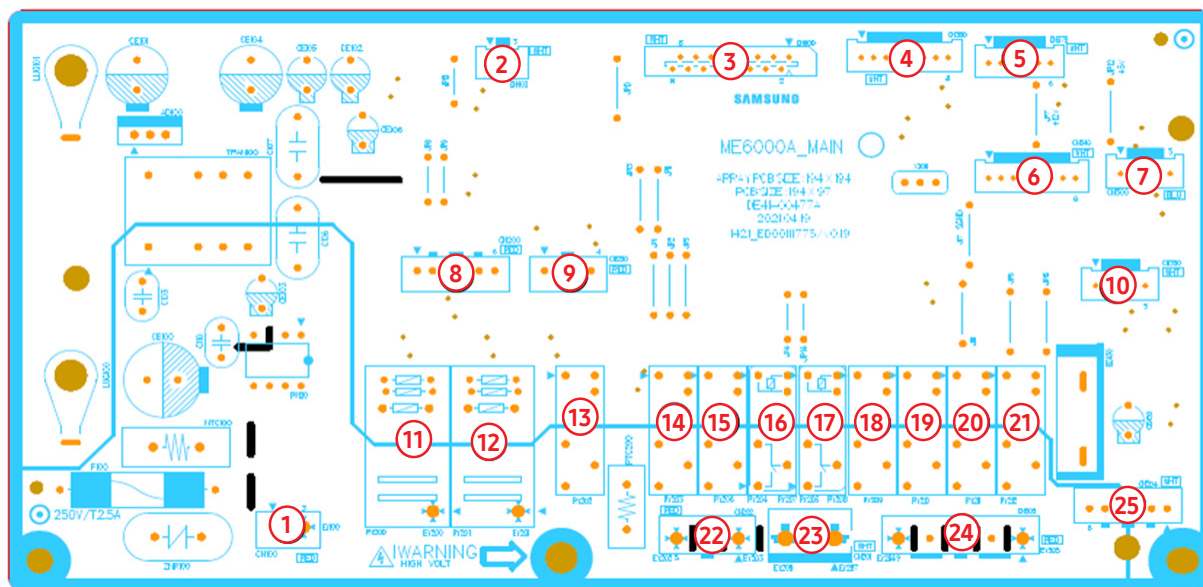
6. PCB Diagrams

6-1 PCB Diagrams (Sub Module)



6. PCB Diagrams

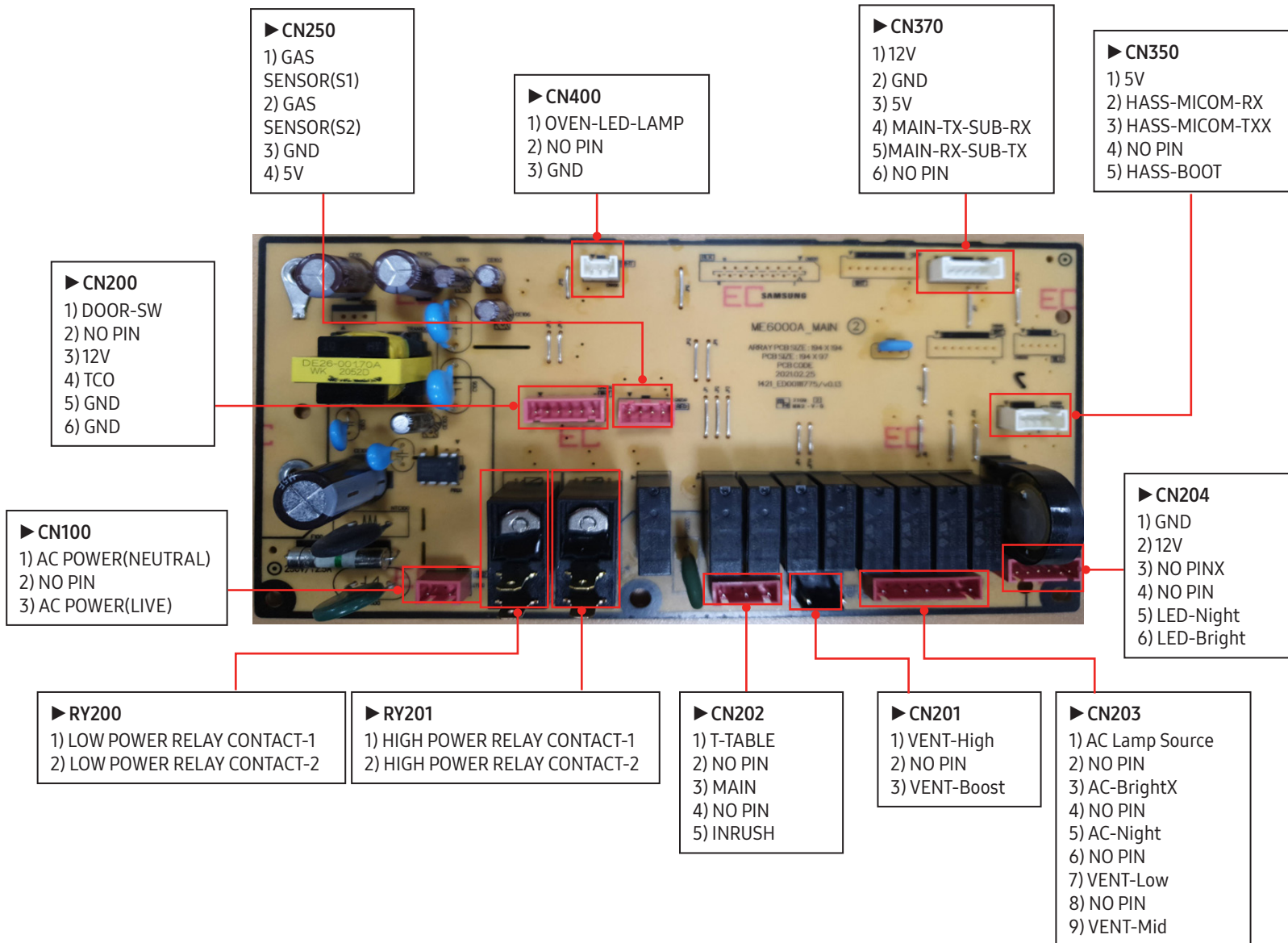
6-2 PCB Diagrams (Main PBA)



No	Part Number	Part name	Function and Rule
1	CN100	Power Connector	Power supply
2	CN400	OVEN LAMP Connector	OVEN LAMP(LED)
3	CN600	Membrane switch Connector	Membrane switch
4	CN380	LED Display Module Connector	LED Display Module
5	CN370	SUB Module Connector	SUB Module
6	CN340	J-Tag Connector	J-Tag Connector
7	CN500	WIFI Connector	WIFI Module
8	CN200	Door Sensor / TCO Sensor Connector	Door Sensor / TCO Sensor
9	CN250	GAS Sensor Connector	Gas Sensor
10	CN350	HASS Connector	HASS
11	RY200	Power High Relay	MW Power Control(High Power)
12	RY201	Power Low Relay	MW Power Control(Low Power)
13	RY202	Inrush Relay	Inrush Electric Current Decrease
14	RY203	Main Relay	Fan, Lamp, T/T Control
15	RY206	Turn Table Relay	T/Table Motor Control
18	RY204/RY207	Vent Boost Relay	Turn on Boost Ventilation
19	RY205/RY208	Vent High Relay	Turn on High Ventilation
20	RY209	Vent Medium Relay	Turn on Medium Ventilation
21	RY210	Vent Low Relay	Turn on Low Ventilation
22	RY211	Night Relay	Turn on Low Light
16	RY212	Bright Relay	Turn on High Light
17	CN202	Relay Control	A terminal for Relay Contact (Inrush, Main, T/Table)
23	CN201	Relay Control	A terminal for Relay Contact (Vent-Boost, Vent-High)
24	CN203	Relay Control	A terminal for Relay Contact (AC Lamp - Bright, Night)
25	CN204	Relay Control	A terminal for Relay Contact (LED Lamp - Bright, Night)

6. PCB Diagrams

6-2 PCB Diagrams (Main PBA)



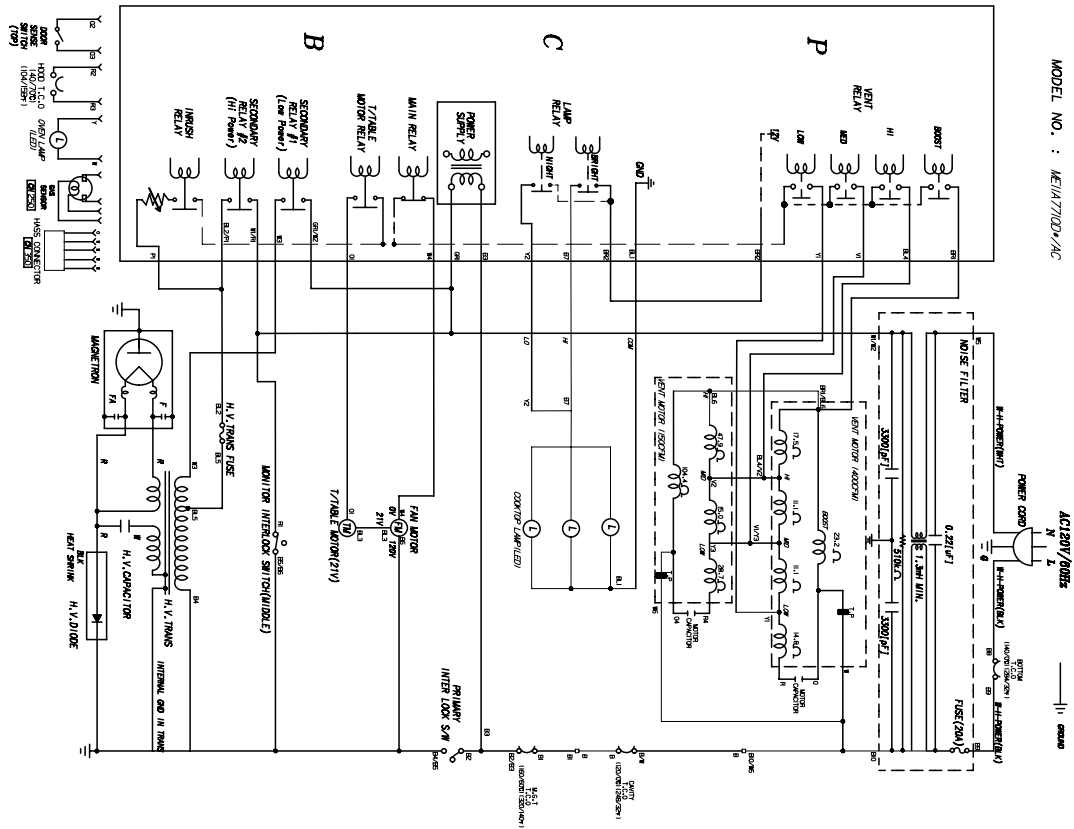
7. Wiring Diagrams

7-1 Wiring Diagrams

SCHEMATIC DIAGRAM

WARNING

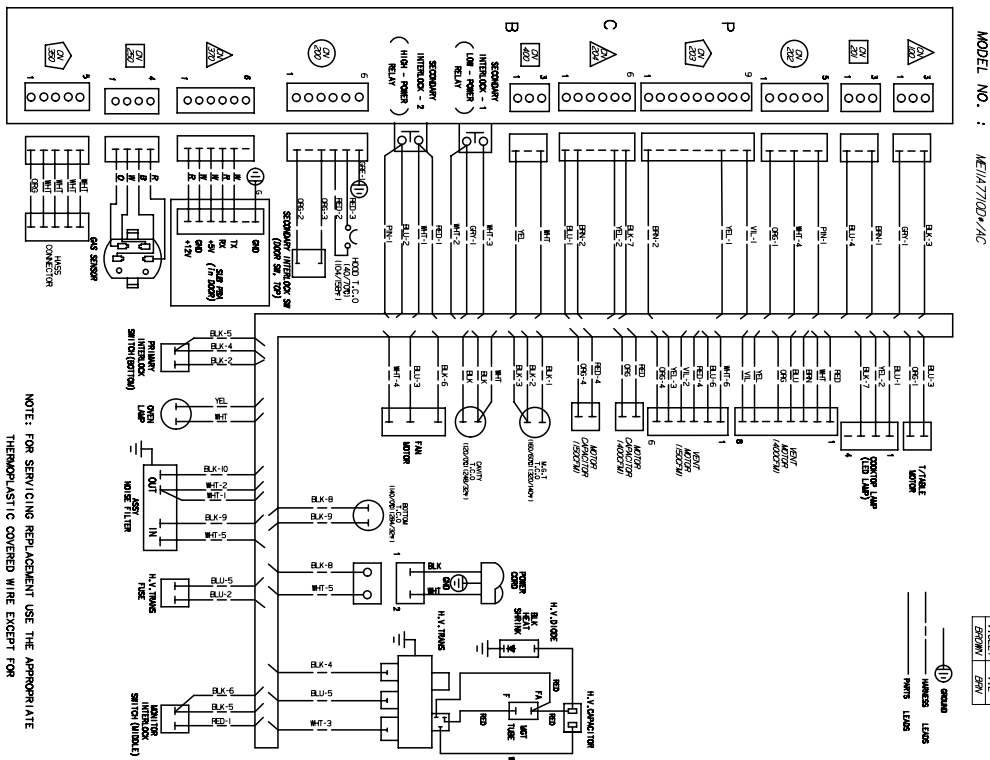
POWER MUST BE DISCONNECTED
BEFORE SERVICING THIS APPLIANCE



WIRING DIAGRAM

WARNING

POWER MUST BE DISCONNECTED
BEFORE SERVICING THIS APPLIANCE



7. Wiring Diagrams

7-1 Wiring Diagrams

SCHEMATIC DIAGRAM

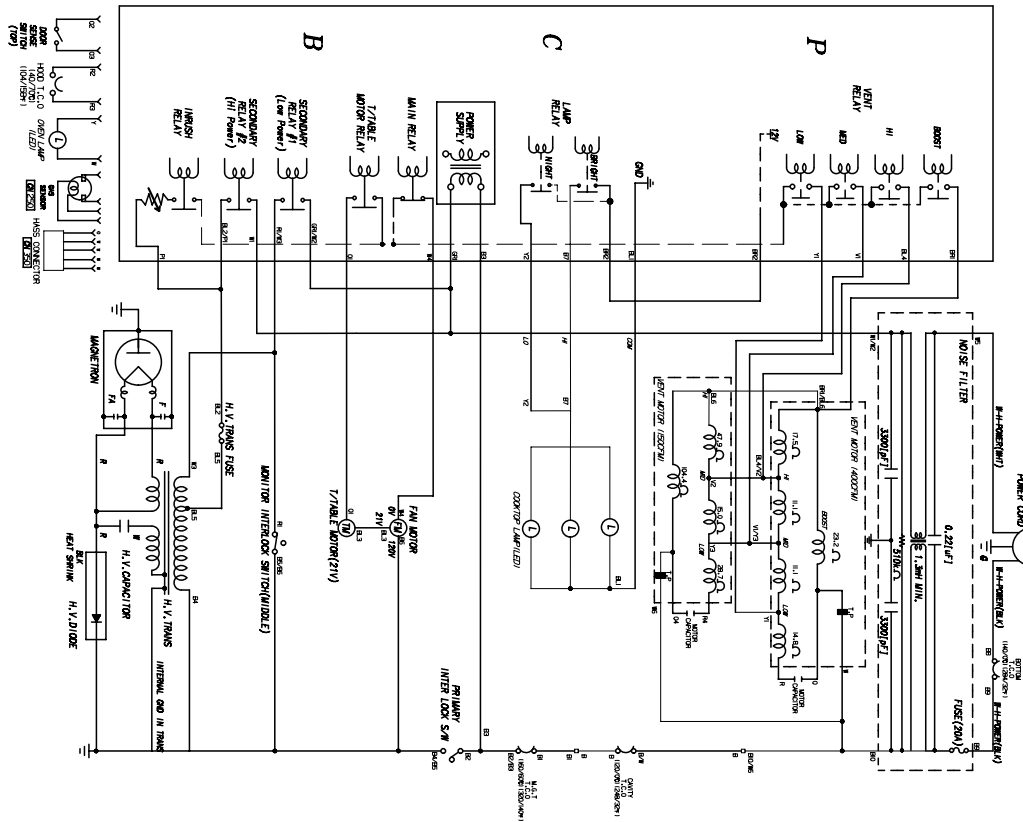
WARNING

POWER MUST BE DISCONNECTED
BEFORE SERVICING THIS APPLIANCE

MODEL NO. : ME7A7700N/AA

AC120V/60Hz

COLOR	SYMBOL
GRAY	GR
WHITE	W
BLACK	B
RED	RED
BLUE	BLU
ORANGE	OR
YELLOW	Y
GREEN	G
PINK	P
VIOLET	V
BROWN	BR



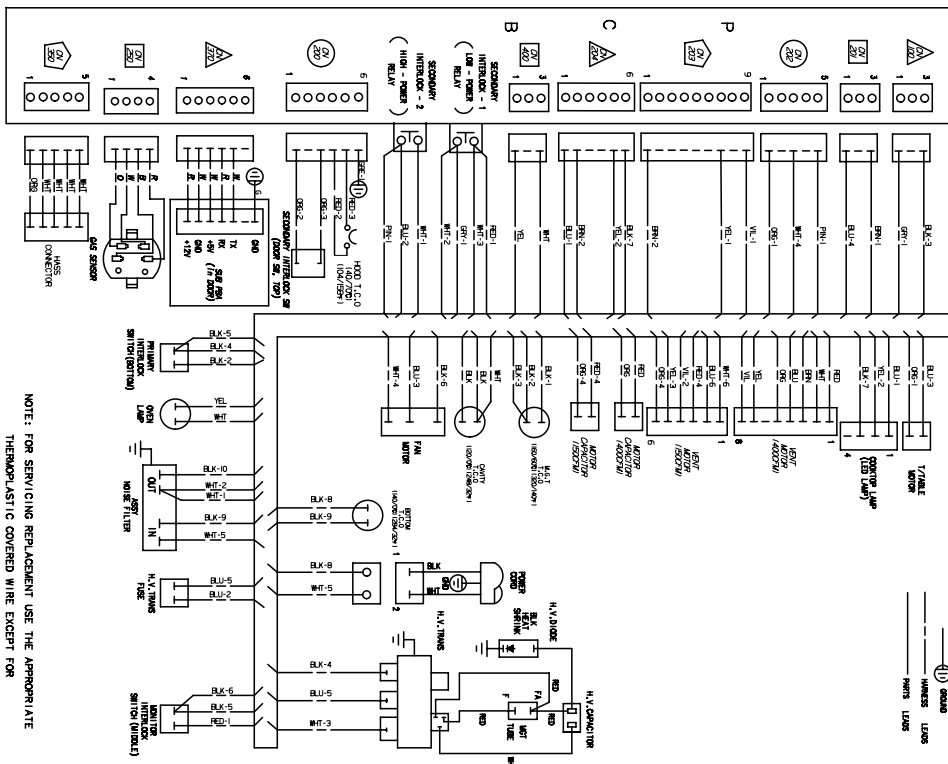
WIRING DIAGRAM

WARNING

POWER MUST BE DISCONNECTED
BEFORE SERVICING THIS APPLIANCE

MODEL NO. : ME7A7700N/AA

COLOR	SYMBOL
GRAY	GR
WHITE	W
BLACK	B
RED	RED
BLUE	BLU
ORANGE	OR
YELLOW	Y
GREEN	G
PINK	P
VIOLET	V
BROWN	BR



NOTE: FOR SERVICING REPLACEMENT USE THE APPROPRIATE
THERMOPLASTIC COVERED WIRE EXCEPT FOR
HIGH VOLTAGE LEADS OR AS NOTED ON SPECIAL LEADS.
Printed in Mexico

7. Wiring Diagrams

7-1 Wiring Diagrams

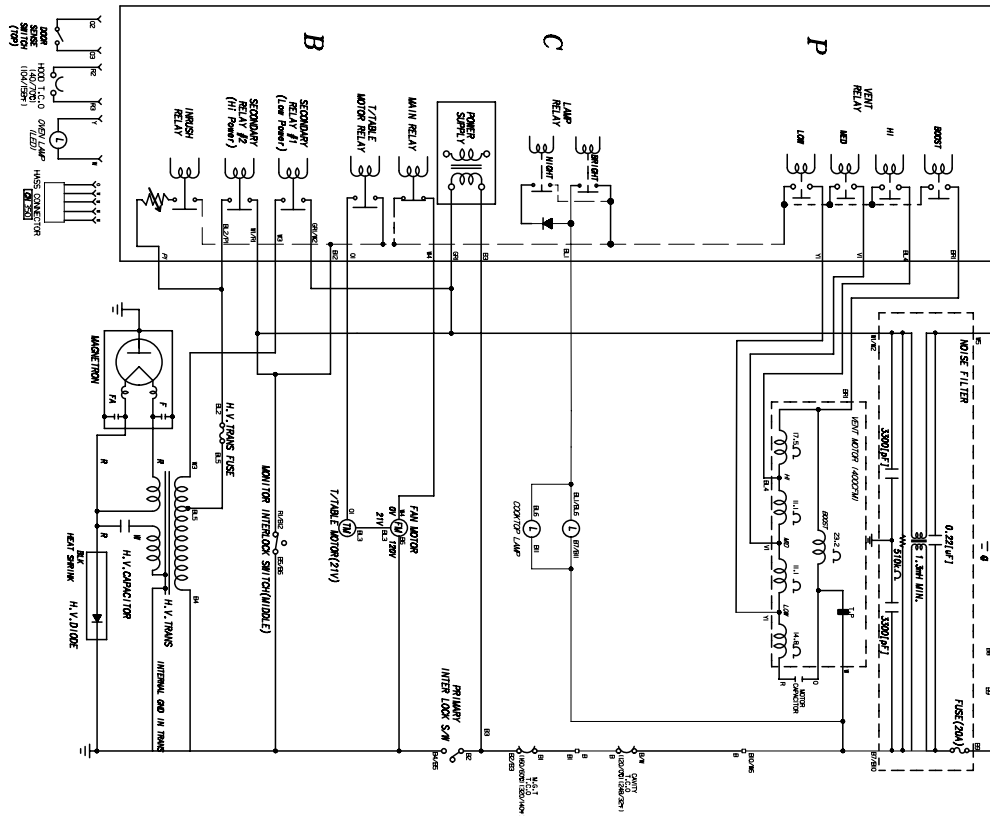
SCHEMATIC DIAGRAM

WARNING
POWER MUST BE DISCONNECTED
BEFORE SERVICING THIS APPLIANCE

MODEL NO. : ME1A7510D/A/C

AC180V/60Hz

COLOR	SYMBOL
GRAY	SW
W/TE	W
BLK/KN	B
BLK/BL	BL
ORANGE	O
YELLOW	Y
GREEN	G
RED	R
BLUE	BL
VIOLET	V
GRAY/WH	SW

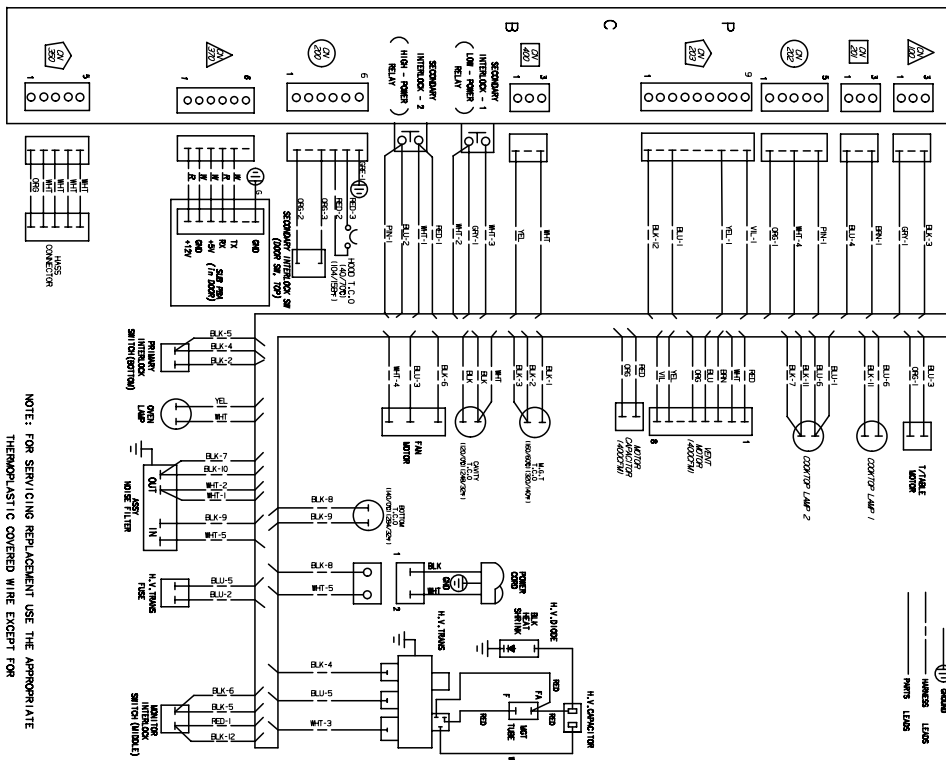


WIRING DIAGRAM

WARNING
POWER MUST BE DISCONNECTED
BEFORE SERVICING THIS APPLIANCE

MODEL NO. : ME1A7510D/A/C

COLOR	SYMBOL
GRAY	SW
W/TE	W
BLK/KN	B
BLK/BL	BL
ORANGE	O
YELLOW	Y
GREEN	G
RED	R
BLUE	BL
VIOLET	V
GRAY/WH	SW



NOTE: FOR SERVICING REPLACEMENT USE THE APPROPRIATE
THERMOPLASTIC COVERED WIRE EXCEPT FOR
HIGH VOLTAGE LEADS OR AS NOTED ON SPECIAL LEADS.
Printed in Malaysia

7. Wiring Diagrams

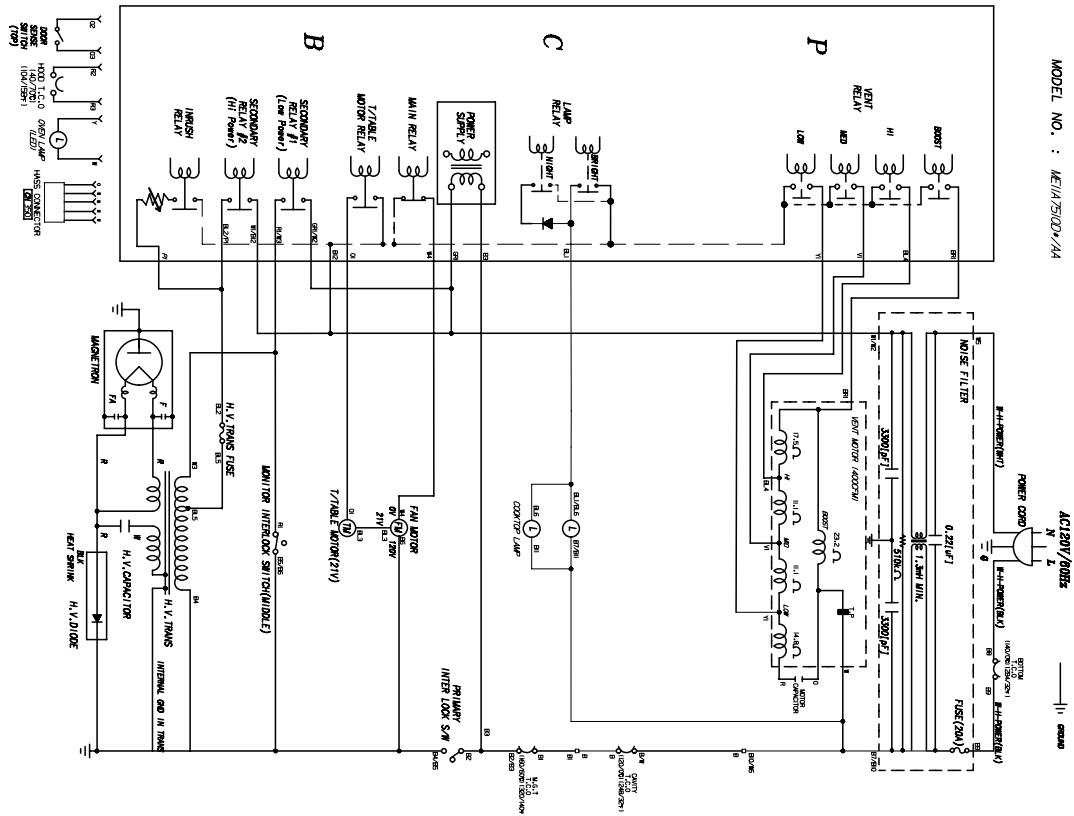
7-1 Wiring Diagrams

SCHEMATIC DIAGRAM

WARNING

POWER MUST BE DISCONNECTED
BEFORE SERVICING THIS APPLIANCE

MODEL NO. : ME1A7510D/A4

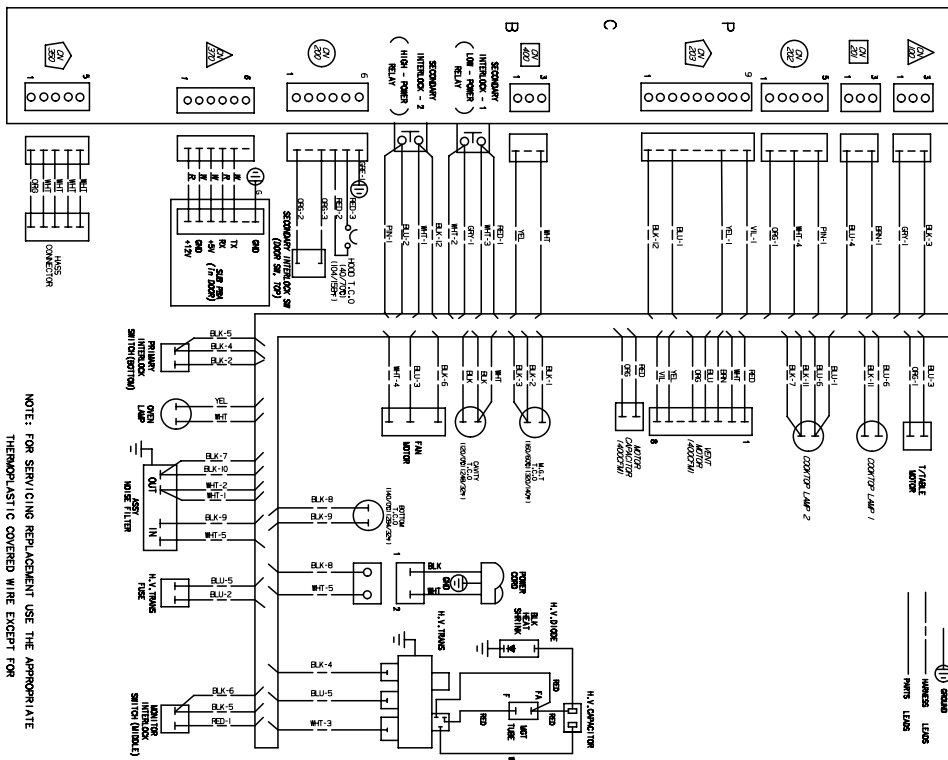


WIRING DIAGRAM

WARNING

POWER MUST BE DISCONNECTED
BEFORE SERVICING THIS APPLIANCE

MODEL NO. : ME1A7510D/A4



NOTE: FOR SERVICING REPLACEMENT USE THE APPROPRIATE
THERMOPLASTIC COVERED WIRE EXCEPT FOR
HIGH VOLTAGE LEADS OR AS NOTED ON SPECIAL LEADS.
Printed in Malaysia



GSPN (GLOBAL SERVICE PARTNER NETWORK)

Area	Web Site
Europe, CIS, Mideast & Africa	gspn1.samsungcsportal.com
Asia	gspn2.samsungcsportal.com
North & Latin America	gspn3.samsungcsportal.com
China	china.samsungportal.com